MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR
LINE DRIVERS AND RECEIVERS

MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon line drivers and receivers for use as voltage comparators and in balanced or unbalanced differential transmission systems which may include party-line transmission systems. Two product assurance classes and a choice of case outline or lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Dual line receiver
02	Dual line receiver (open collector output)
03	Dual differential line driver (two-state output)
04	Dual differential line receiver
05	Dual differential line driver (three-state output with high-impedance off-state)
06	Dual differential line driver (three-state output with high-impedance off-state) with output diode clamping
07	Dual differential line driver (three-state output with high-impedance off-state)

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
B	F-3 (14-lead, 3/16" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package
2	C-2 (20-terminal, .350" x .350"), square chip carrier package

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5962

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1.3 Absolute maximum ratings.

1.3.1 Device types 01 and 02:

Maximum power dissipation (P_D)	550 mW
Positive supply voltage ($+V_{CC}$)	+7.0 V dc
Negative supply voltage ($-V_{CC}$)	-7.0 V dc
Differential input voltage	-6.0 V dc to +6.0 V dc
Common-mode input voltage	-5.0 V dc to +5.0 V dc
Strobe input voltage	+5.5 V dc
Lead temperature (soldering, 10 seconds)	+300°C
Junction temperature (T_J)	+175°C
Storage temperature range	-65°C to +150°C
Thermal resistance:	
Junction-to-case (θ_{JC})	28°C/W for dual-in-line package 22°C/W for flat package 20°C/W for square chip carrier package
Junction-to-ambient (θ_{JA})	91°C/W for dual-in-line package 119°C/W for flat package 125°C/W for square chip carrier package

1.3.2 Device types 03 and 05:

Supply voltage range	-0.5 V dc to +7.0 V dc
Input data voltage range	-1.5 V dc at -12 mA to +5.5 V dc
Open collector output voltage range	-1.5 V dc at -40 mA to V dc
Maximum power dissipation (P_D)	400 mW
Lead temperature (soldering, 10 seconds)	+300°C
Junction temperature (T_J)	+175°C
Storage temperature range	-65°C to +150°C
Thermal resistance:	
Junction-to-case (θ_{JC})	28°C/W for dual-in-line package 22°C/W for flat package 20°C/W for square chip carrier package
Junction-to-ambient (θ_{JA})	91°C/W for dual-in-line package 119°C/W for flat package 125°C/W for square chip carrier package

1.3.3 Device type 04:

Supply voltage range	-0.5 V dc to +7.0 V dc
Input data voltage range	-20 V dc to +20 V dc
Input strobe voltage range	-0.5 V dc to +5.5 V dc
Maximum power dissipation (P_D)	400 mW
Lead temperature (soldering, 10 seconds)	+300°C
Junction temperature (T_J)	+175°C
Storage temperature range	-65°C to +150°C
Thermal resistance:	
Junction-to-case (θ_{JC})	28°C/W for dual-in-line package 22°C/W for flat package 20°C/W for square chip carrier package
Junction-to-ambient (θ_{JA})	91°C/W for dual-in-line package 119°C/W for flat package 125°C/W for square chip carrier package

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1.3.4 Device types 06 and 07:

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -12 mA to +5.5 V dc
Input strobe voltage range	-0.5 V dc to +5.5 V dc
Maximum power dissipation (P_D)	220 mW
Lead temperature (soldering, 10 seconds)	+300°C
Junction temperature (T_J)	+175°C
Storage temperature range	-65°C to +150°C
Thermal resistance:	
Junction-to-case (θ_{JC})	28°C/W for dual-in-line package 22°C/W for flat package 20°C/W for square chip carrier package
Junction-to-ambient (θ_{JA})	91°C/W for dual-in-line package 119°C/W for flat package 125°C/W for square chip carrier package

1.4 Recommended operating conditions.

1.4.1 Device types 01 and 02:

Supply voltage ($+V_{CC}$)	+4.5 V dc minimum to +5.5 V dc maximum
Supply voltage ($-V_{CC}$)	-4.5 V dc minimum to -5.5 V dc maximum
Output sink current	-16 mA maximum
Differential input voltage	-5 V dc minimum to +5 V dc maximum
Common-mode input voltage	-3 V dc minimum to +3 V dc maximum
Input voltage range	-5 V dc minimum to +3 V dc maximum
Ambient operating temperature range	-55°C minimum to +125°C maximum

1.4.2 Device types 03 and 05:

Supply voltage	4.5 V dc minimum to 5.5 V dc maximum
Minimum high level input voltage	2.0 V dc
Maximum low level input voltage	0.8 V dc
Ambient operating temperature range	-55°C to +125°C

1.4.3 Device type 04:

Supply voltage	4.5 V dc minimum to 5.5 V dc maximum
Common-mode input voltage (data inputs)	-15 V dc to +15 V dc
Minimum high level input voltage (strobe input)	2.0 V dc
Maximum low level input voltage (strobe input)	0.8 V dc
Ambient operating temperature range	-55°C to +125°C

1.4.4 Device types 06 and 07:

Supply voltage	4.5 V dc minimum to 5.5 V dc maximum
Minimum high level input voltage	2.0 V dc
Maximum low level input voltage	0.8 V dc
Ambient operating temperature range	-55°C to +125°C

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standard, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Logic diagram and terminal connections. The logic diagram and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3 herein.

3.3 Lead material and finish. Lead material and finish shall be in accordance with MIL-M-38510.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range, unless otherwise specified.

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TABLE I. Electrical performance characteristics.

Characteristic	Symbol	Conditions	Limits		Unit
			Min	Max	
Device types 01 and 02					
High level input current into 1A or 2A	I_{IH1}	$+V_{CC}$ = +5.5 V; $-V_{CC}$ = -5.5 V; V_{ID} = 0.5 V; V_{IC} = -3 V to +3 V		75	µA
High level input current into S	I_{IH2}	$+V_{CC}$ = +5.5 V; $-V_{CC}$ = -5.5 V; V_{IN} = 2.4 V		80	µA
	I_{IH3}	$+V_{CC}$ = +5.5 V; $-V_{CC}$ = -5.5 V; V_{IN} = 5.5 V		2	mA
High level input current into 1G or 2G	I_{IH4}	$+V_{CC}$ = +5.5 V; $-V_{CC}$ = -5.5 V; V_{IN} = 2.4 V		40	µA
	I_{IH5}	$+V_{CC}$ = +5.5 V; $-V_{CC}$ = -5.5 V; V_{IN} = 5.5 V		100	µA
Low level input current into 1A or 2A	I_{IL1}	$+V_{CC}$ = +5.5 V; $-V_{CC}$ = -5.5 V; V_{ID} = -2 V; V_{IC} = -3 V to +3 V		-10	µA
Low level input current into 1G or 2G	I_{IL2}	$+V_{CC}$ = +5.5 V; $-V_{CC}$ = -5.5 V; V_{IN} = 0.4 V	-0.7	-1.6	mA
Low level input current into S	I_{IL3}	$+V_{CC}$ = +5.5 V; $-V_{CC}$ = -5.5 V; V_{IN} = 0.4 V	-1.4	-3.2	mA
High level output voltage	V_{OH}	$+V_{CC}$ = +4.5 V; $-V_{CC}$ = -4.5 V; I_{OH} = -400 µA; V_{IC} = -3 V to +3 V (type 01 only)	2.4		V
Low level output voltage	V_{OL}	$+V_{CC}$ = +4.5 V; $-V_{CC}$ = -4.5 V; I_{OL} = 16 mA; V_{IC} = -3 V to +3 V		0.4	V
Short-circuit output current	I_{OS}	$+V_{CC}$ = +5.5 V; $-V_{CC}$ = -5.5 V 1/ (type 01 only)	-18	-70	mA
High level output current	I_{OH}	$+V_{CC}$ = +4.5 V; $-V_{CC}$ = -4.5 V; V_{OH} = 5.5 V (type 02 only)		250	µA
High level supply current from $+V_{CC}$	I_{CCH+}	$+V_{CC}$ = +5.5 V; $-V_{CC}$ = -5.5 V; V_{ID} = 25 mV		30	mA
High level supply current from $-V_{CC}$	I_{CCH-}	$+V_{CC}$ = +5.5 V; $-V_{CC}$ = -5.5 V; V_{ID} = 25 mV		-15	mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions	Limits		Unit
			Min	Max	
Device type 01					
Propagation delay to high level (inputs A and B to output)	t_{PLH1}	$+V_{CC}$ = +5.0 V; $-V_{CC}$ = -5.0 V; R_L = 390Ω; C_L = 50 pF		40	ns
Propagation delay to low level (inputs A and B to output)	t_{PHL1}			35	ns
Propagation delay to high level (strobe inputs G or S to output)	t_{PLH2}			30	ns
Propagation delay to low level (strobe inputs G or S to output)	t_{PHL2}			25	ns
Device type 02					
Propagation delay to high level (inputs A and B to output)	t_{PLH1}	$+V_{CC}$ = 5.0 V; $-V_{CC}$ = -5.0 V; R_L = 390Ω; C_L = 50 pF		45	ns
Propagation delay to low level (inputs A and B to output)	t_{PHL1}			35	ns
Propagation delay to high level (strobe inputs G or S to output)	t_{PLH2}			40	ns
Propagation delay to low level (strobe inputs G or S to output)	t_{PHL2}			25	ns
Device type 03					
High level output voltage	V_{OH1}	V_{CC} = 4.5 V; V_{IN} = 0.8 V or 2.0 V; I_{OH} = -10 mA	2.4		V
	V_{OH2}	V_{CC} = 4.5 V; V_{IN} = 0.8 V or 2.0 V; I_{OH} = -20 mA	2.4 2/		V
Low level output voltage	V_{OL}	V_{CC} = 4.5 V; V_{IN} = 0.8 V or 2.0 V; I_{OL} = 40 mA		0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V; I_{IC} = -12 mA; T_A = +25°C		-1.5	V
Output clamp voltage	V_{OC}	V_{CC} = 5.5 V; I_{OLC} = -40 mA; T_A = +25°C		-1.5	V

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions	Limits		Unit
			Min	Max	
Device type 03 - Continued					
Low level input current	I_{IL}	V_{CC} = 5.5 V; V_{IN} = 0.4 V	-0.7	-1.6	mA
High level input current (inputs A, B, and C)	I_{IH1}	V_{CC} = 5.5 V; V_{IN} = 2.4 V		40	µA
	I_{IH2}	V_{CC} = 5.5 V; V_{IN} = 5.5 V		100	µA
High level input breakdown voltage	BV_{IN}	V_{CC} = 5.5 V; I_{IN} = 1 mA	5.5		V
Short-circuit output current	I_{OS}	V_{CC} = 5.5 V; V_{OUT} = 0 V 1/ T_A = +25°C, +125°C	-40	-130	mA
		T_A = -55°C	-40	-155	mA
Output leakage current	I_{CEX}	V_{CC} = 5.5 V; V_{CEX} = 12 V		200	µA
Power supply current	I_{CC}	V_{CC} = 5.5 V		50	mA
Propagation delay to high level (input to 1Z or 2Z)	t_{PLH1}	V_{CC} = 5.0 V; C_L = 50 pF		32	ns
Propagation delay to low level (input to 1Z or 2Z)	t_{PHL1}			30	ns
Propagation delay to high level (input to 1Y or 2Y)	t_{PLH2}			30	ns
Propagation delay to low level (input to 1Y or 2Y)	t_{PHL2}			32	ns
Device type 04					
High level output voltage	V_{OH}	V_{CC} = 4.5 V; I_{OH} = -5 mA	2.4		V
Low level output voltage	V_{OL}	V_{CC} = 4.5 V; I_{OL} = 16 mA		0.4	V
Supply current (both receivers)	I_{CC}	V_{CC} = 5.5 V; V_{IC} = 0 V; V_{DIFF} = 0.5 V		50	mA
Input clamp voltage (strobe)	V_{IC}	V_{CC} = 4.5 V; I_{IC} = -12 mA; T_A = +25°C		-1.5	V

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions	Limits		Unit
			Min	Max	
Device type 04 - Continued					
High level input breakdown voltage (strobe)	BV_{IN}	V_{CC} = 5.5 V; I_{IN} = 1 mA	5.5		V
Low level input current (data input)	I_{IL1}	V_{CC} = 5.5 V; V_{IN} = 0.4 V; other input = 5.5 V	-0.2	-0.9	mA
Low level input current (strobe)	I_{IL2}	V_{CC} = 5.5 V; V_{IN} = 0.4 V; V_{DIFF} = 0.5 V	-1.0	-2.4	mA
Low level input current (response control)	I_{IL3}	V_{CC} = 5.5 V; V_{IN} = 0.4 V; V_{DIFF} = 0.5 V	-1.2	-4.1	mA
High level input current (strobe)	I_{IH}	V_{CC} = 5.5 V; V_{IN} = 4.5 V; V_{DIFF} = -0.5 V		5.0	µA
Input resistor	R_{IN}	T_A = +25°C; V_{CC} = 5.0 V; V_{IN} = 1.0 V	77	167	Ω
Output leakage current	I_{CEX}	V_{CC} = 4.5 V; V_{OUT} = 12 V		200	µA
Output short circuit current	I_{OS}	V_{CC} = 5.5 V 1/	-15	-80	mA
Differential input threshold voltage	V_{TH}	V_{CM} = 0 V		500	mV
		V_{CM} = ±15 V		2.0	V
Propagation delay to low level (inputs A and B to output)	t_{PHL1}	V_{CC} = 5.0 V; C_L = 50 pF; R_L = 390Ω for t_{PHL}; R_L = 3,900Ω for t_{PLH}		60	ns
Propagation delay to high level (inputs A and B to output)	t_{PLH1}			60	ns
Propagation delay to low level (strobe to output)	t_{PHL2}			25	ns
Propagation delay to high level (strobe to output)	t_{PLH2}			25	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions	Limits		Unit
			Min	Max	
Device type 05					
High level output voltage	V_{OH1}	V_{CC} = 4.5 V; I_{OH} = -10 mA	2.4		V
	V_{OH2}	V_{CC} = 4.5 V; I_{OH} = -40 mA; T_A = +25°C	2.0		V
Low level output voltage	V_{OL}	V_{CC} = 4.5 V; I_{OL} = 40 mA		0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V; I_{IC} = -12 mA; T_A = +25°C		-1.5	V
Output clamp voltage	V_{OC}	V_{CC} = 5.5 V; I_O = -40 mA		-1.5	V
Low level input current (inputs A, B, and C)	I_{IL1}	V_{CC} = 5.5 V; V_{IN} = 0.4 V	-0.7	-1.6	mA
Low level input current (input D)	I_{IL2}	V_{CC} = 5.5 V; V_{IN} = 0.4 V	-1.4	-3.2	mA
High level input current (inputs A, B, and C)	I_{IH1}	V_{CC} = 5.5 V; V_{IN} = 2.4 V		40	µA
	I_{IH2}	V_{CC} = 5.5 V; V_{IN} = 5.5 V		100	µA
High level input current (input D)	I_{IH3}	V_{CC} = 5.5 V; V_{IN} = 2.4 V		80	µA
	I_{IH4}	V_{CC} = 5.5 V; V_{IN} = 5.5 V		200	µA
Off-state open collector output leakage current (outputs YP and ZP)	I_{O1}	V_{CC} = 5.5 V; V_O = 12 V		200	µA
Inhibited-state output leakage current (sink tied to pull-up)	I_{O2}	V_{CC} = 5.5 V; V_O = 0.0 V		-120	µA
	I_{O3}	V_{CC} = 5.5 V; V_O = 0.4 V		±80	µA
	I_{O4}	V_{CC} = 5.5 V; V_O = 2.4 V		±80	µA
	I_{O5}	V_{CC} = 5.5 V; V_O = 5.0 V		200	µA
Short-circuit output current	I_{OS}	V_{CC} = 5.5 V; V_O = 0.0 V	-40	-120	mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions	Limits		Unit
			Min	Max	
Device type 05 - Continued					
Supply current (both drivers)	I_{CC1}	V_{CC} = 5.5 V; V_{IN} = 0.0 V; T_A = +25°C		65	mA
	I_{CC2}	V_{CC} = 7.0 V; V_{IN} = 0.0 V; T_A = +25°C		85	mA
Propagation delay time (low-to-high level output)	t_{PLH} (1 and 2)	V_{CC} = 5.0 V; C_L = 50 pF		30	ns
Propagation delay time (high-to-low level output)	t_{PHL} (1 and 2)	V_{CC} = 5.0 V; C_L = 50 pF		30	ns
Propagation delay time (disabled to high level output)	t_{PZH}	V_{CC} = 5.0 V; C_L = 50 pF; R_L = 180Ω; V_L = 0.0 V		25	ns
Propagation delay time (high level to disabled output)	t_{PHZ}	V_{CC} = 5.0 V; C_L = 50 pF; R_L = 180Ω; V_L = 0.0 V		35	ns
Propagation delay time (disabled to low level output)	t_{PZL}	V_{CC} = 5.0 V; C_L = 50 pF; R_L = 250Ω; V_L = 5.0 V		35	ns
Propagation delay time (low level to disabled output)	t_{PLZ}	V_{CC} = 5.0 V; C_L = 50 pF R_L = 250Ω; V_L = 5.0 V		45	ns
Device types 06 and 07					
High level output voltage	V_{OH}	V_{CC} = 4.5 V; V_{IL} = 0.8 V; V_{IH} = 2.0 V; I_{OH} = -.2 mA; V_{IL} = 0.7 V at +125°C	2.4		V
Low level output voltage	V_{OL}	V_{CC} = 4.5 V; V_{IL} = 0.8 V; V_{IH} = 2.0 V; I_{OL} = 40 mA; V_{IL} = 0.7 V at +125°C		0.5	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V; I_{IC} = -12 mA; T_A = +25°C		-1.5	V
Output clamp voltage	V_{OC}	V_{CC} = 5.5 V; I_{OC} = -12 mA; T_A = +25°C 3/		-1.5	V
Low level input current	I_{IL}	V_{CC} = 5.5 V; V_{IN} = 0.4 V	-0.7	-1.6	mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions	Limits		Unit
			Min	Max	
Device types 06 and 07 - Continued					
High level input current	I_{IH1}	V_{CC} = 5.5 V; V_{IH} = 2.4 V		40	µA
	I_{IH2}	V_{CC} = 5.5 V; V_{IH} = 5.5 V		100	µA
Output short circuit current	I_{OS}	V_{CC} = 5.5 V 1/	-40	-120	mA
Supply current	I_{CC}	V_{CC} = 5.5 V		90	mA
Inhibited-state output leakage current	I_{O1}	V_{CC} = 5.5 V; V_{OUT} = 2.4 V	-160	1.0	µA
	I_{O2}	V_{CC} = 5.5 V; V_{OUT} = 0.4 V	-160	1.0	µA
Propagation delay time (low-to-high level output)	t_{PLH}	V_{CC} = 5.0; C_L = 50 pF	2	28	ns
Propagation delay time (high-to-low level output)	t_{PHL}	V_{CC} = 5.0; C_L = 50 pF	2	28	ns
Propagation delay time (disabled to high level output)	t_{PZH}	V_{CC} = 5.0; C_L = 50 pF	2	32	ns
Propagation delay time (high level to disabled output)	t_{PHZ}	V_{CC} = 5.0; C_L = 50 pF	2	34	ns
Propagation delay time (disabled to low level output)	t_{PZL}	V_{CC} = 5.0; C_L = 50 pF	2	32	ns
Propagation delay time (low level to disabled output)	t_{PLZ}	V_{CC} = 5.0; C_L = 50 pF	2	36	ns

1/ Not more than one output should be shorted at a time.
2/ 2.2 V at -55°C.
3/ Type 06 only.

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3.5 Rebonding. Rebonding shall be in accordance with MIL-M-38510.

3.6 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004) 1/	1,2,3,9	1,2,3,9
Group A test requirements (method 5005)	1,2,3,9, 10,11	1,2,3,9
Group C end-point electrical and group B, class S parameters (method 5005)	1,2,3	1,2,3
Additional electrical subgroups for group C periodic inspections	Not applicable	10,11
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

1/ PDA applies to subgroup 1 (see 4.2.d).

3.7 Marking. Marking shall be in accordance with MIL-M-38510.

3.7.1 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.7.2 Correctness of indexing and markings. All devices shall be subjected to the final electrical tests specified in table II after part marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 53 (see MIL-M-38510, appendix E).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Class S devices: Test condition D, using the circuit shown on figure 3.

(2) Class B devices: Test condition D for device types 01 and 02, test condition B for device types 03, 04, 05, 06, and 07, using the circuit shown on figure 3, or equivalent.

b. Reverse bias burn-in (method 1015 of MIL-STD-883). This screen test shall apply to class S devices only using the circuit shown on figure 3.

c. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

d. The percent defective allowable (PDA) for class S and class B devices shall be as specified in MIL-M-38510, based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, 6, 7, and 8 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows.

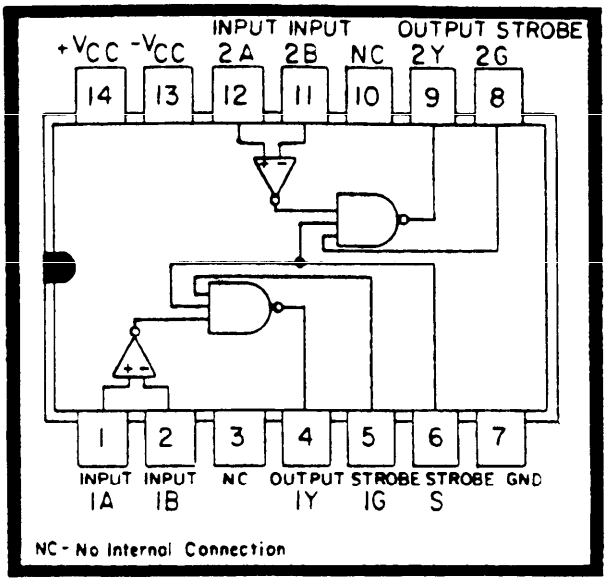
a. End-point electrical parameters shall be as specified in table II herein.

b. Life test for class S (method 1005 of MIL-STD-883). The circuit shown on figure 3 shall be used with alternate burn-in conditions.

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Device types 01 and 02

Cases A, B, C, and D



Device type 03

Cases E and F

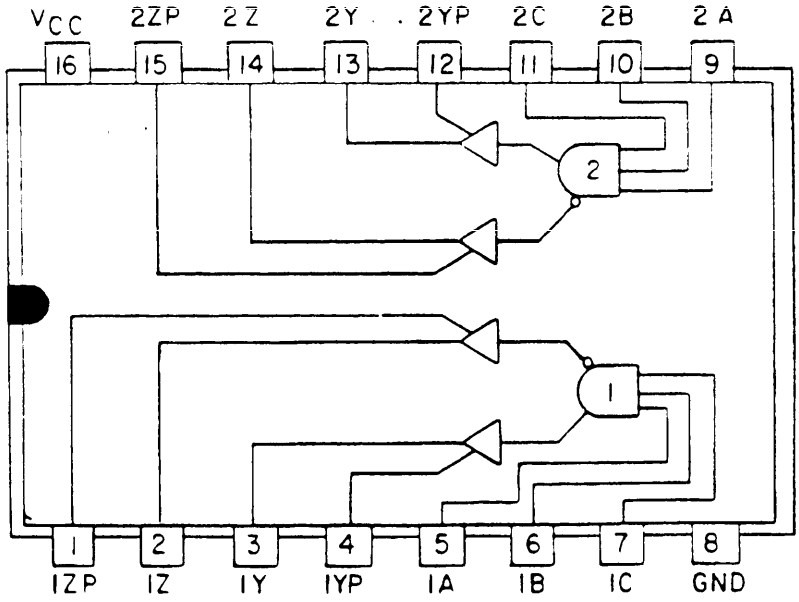
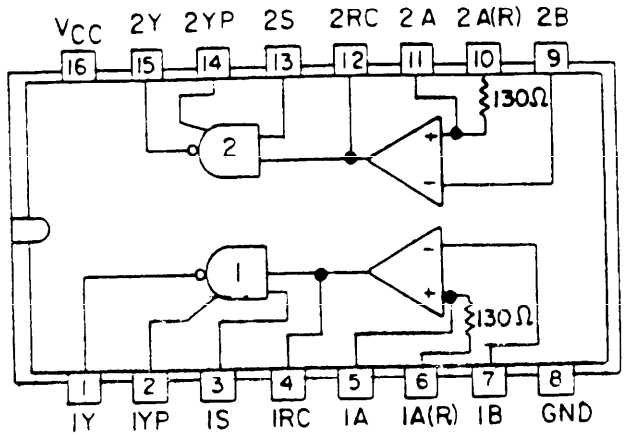


FIGURE 1. Logic diagram and terminal connections.

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Device type 04

Cases E and F



Device type 04

Case 2

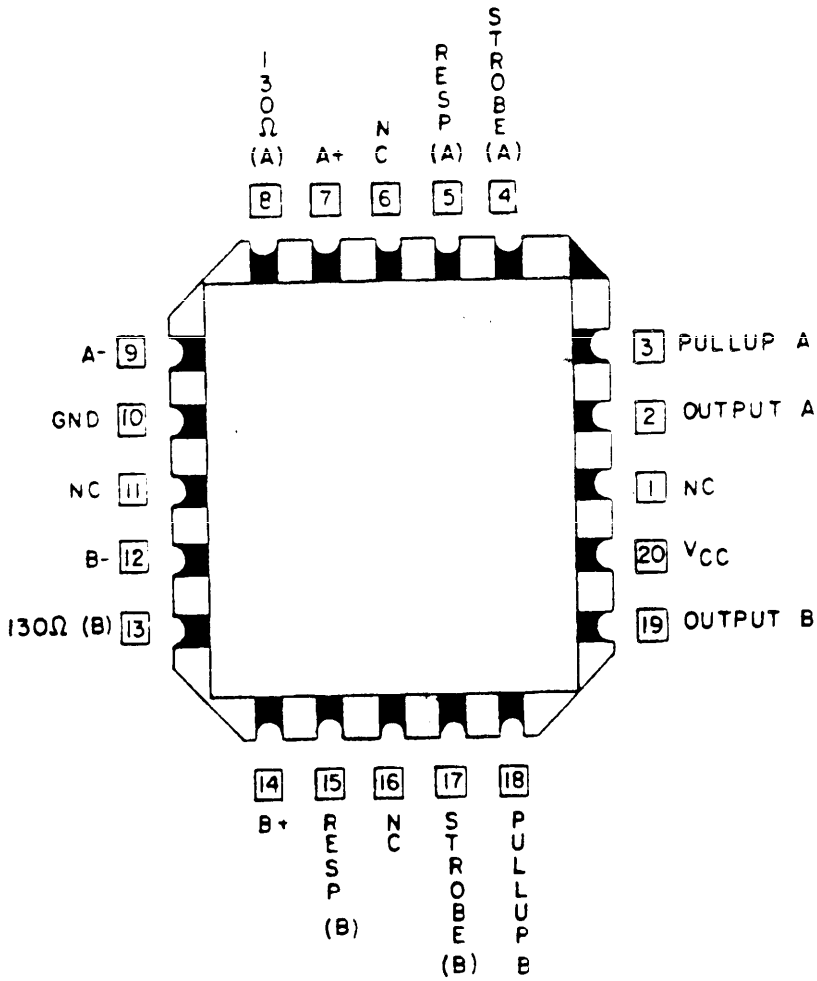
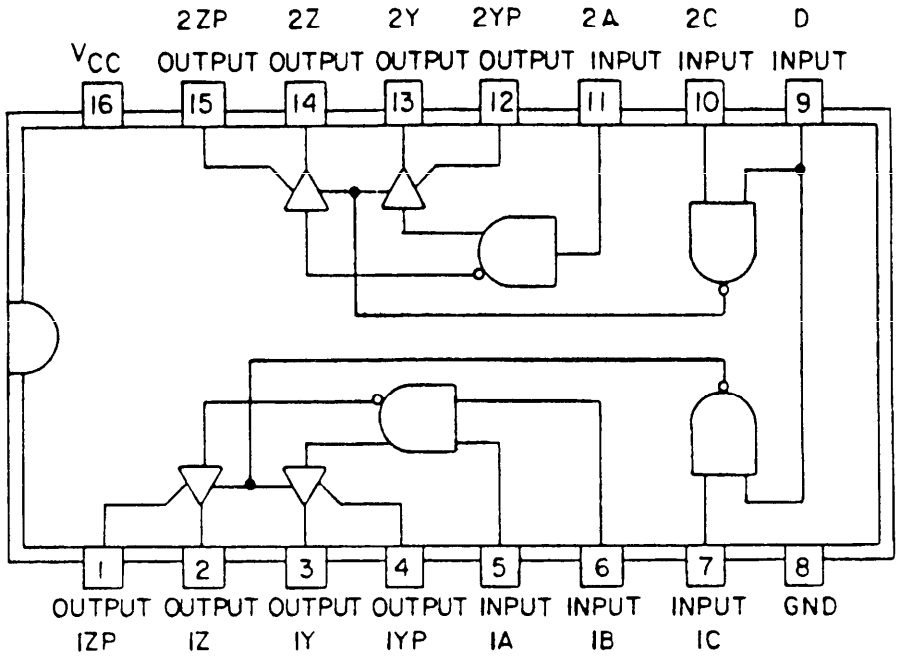


FIGURE 1. Logic diagram and terminal connections - Continued.

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Device type 05

Cases E and F



Device types 06 and 07

Cases E and F

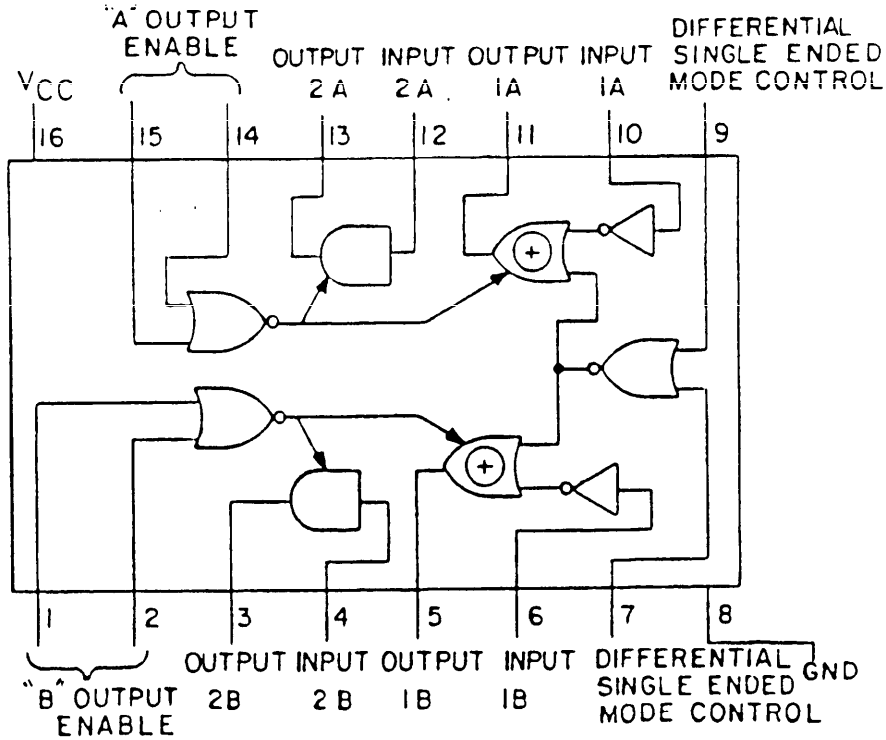


FIGURE 1. Logic diagram and terminal connections - Continued.

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Device types 01 and 02

Differential inputs A-B	Strobes G	Strobes S	Output Y
$V_{ID} \geq 25$ mV	L or H	L or H	H
-25 mV < V_{ID} < 25 mV	L or H	L	H
	L	L or H	H
	H	H	Indeterminate
$V_{ID} \leq -25$ mV	L or H	L	H
	L	L or H	H
	H	H	L

Device type 03

Inputs (1 or 2) A	B	C	Outputs 1Z and 1ZP or 2Z and 2ZP	1Y and 1YP or 2Y and 2YP
L	L	L	H	L
H	L	L	H	L
L	H	L	H	L
H	H	L	H	L
L	L	H	H	L
H	L	H	H	L
L	H	H	H	L
H	H	H	L	H

FIGURE 2. Truth tables and logic equations.

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Device type 04

Inputs			Outputs
Differential inputs		Strobe	
A (1 or 2)	B (1 or 2)	S (1 or 2)	YP/Y (1 or 2) 1/
X 2/	X 2/	L	H
GND	GND	H	Indeterminate
GND	0.5 V	H	H
0.5 V	GND	H	L
0.5 V	0.5 V	H	Indeterminate

1/ Y and YP connected.
2/ X = Don't care.

Device type 05

Inputs				Outputs			
A 1 or 2	B 1 (only)	C 1 or 2	D	ZP	Z	Y	YP
L	L or H	H	H	H	Hi Z	L	Hi Z
L or H	L	H	H	H	Hi Z	L	Hi Z
H	H	H	H	Hi Z	L	Hi Z	H
L or H	L or H	L	L or H	Hi Z	Hi Z	Hi Z	Hi Z
L or H	L or H	L or H	L	Hi Z	Hi Z	Hi Z	Hi Z

Device types 06 and 07

Inputs						Outputs	
Enable		D	SE	MC	1	2	1	2
L	L	L	L	L	L	L	L	L
L	L	L	L	H	H	H	H	H
L	L	X 1	H	H	H	L	H
L	L	X	H	L	L	H	L
L	L	H	X	H	H	L	H
L	L	H	X	L	L	H	L
H	X	X	X	X	X	Hi Z	Hi Z
X	H	X	X	X	X	Hi Z	Hi Z

1 X = Don't care.

FIGURE 2. Truth tables and logic equations - Continued.

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Device types 01 and 02

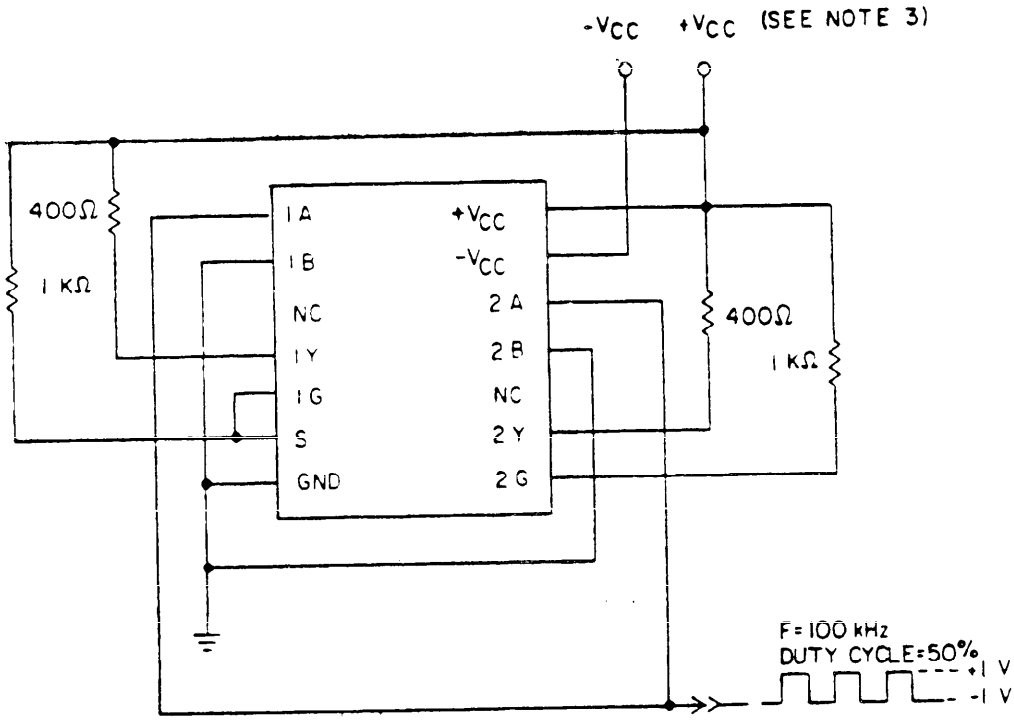
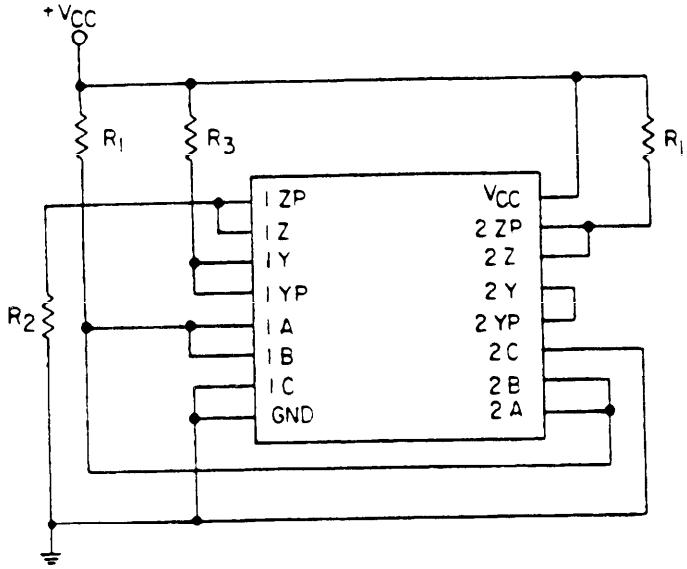


FIGURE 3. Burn-in and life-test circuit.

NOTES:
1. All resistors are 1/4 watt, ±5 percent.
2. All clock pulses are typically 0 V to +1.0 V, 50 percent duty cycle.
3. $+V_{CC}$ shall be a minimum of +5.0 V when measured at the $+V_{CC}$ terminal of the device and $-V_{CC}$ shall be a minimum of -5.0 V when measured at the $-V_{CC}$ terminal of the device.

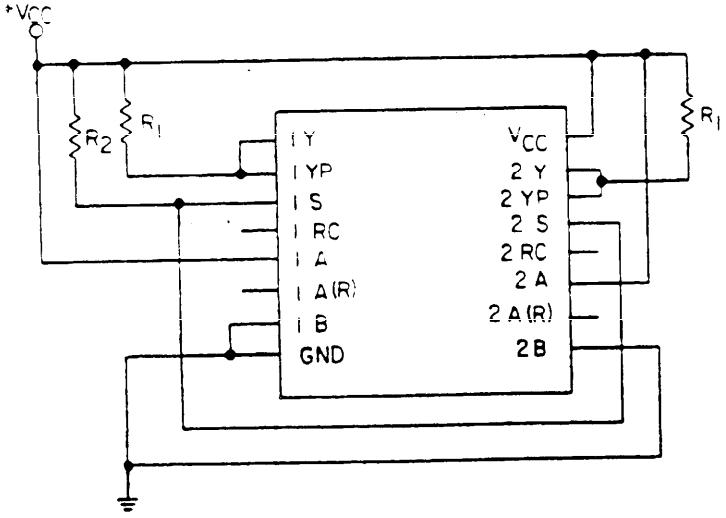
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Device type 03



T_A = 125° C
R_1 = 1000Ω ±5%
R_2 = 240Ω ±5%
R_3 = 110Ω ±5%

Device type 04



T_A = 125° C
R_1 = 510Ω ±5%
R_2 = 1000Ω ± 5%

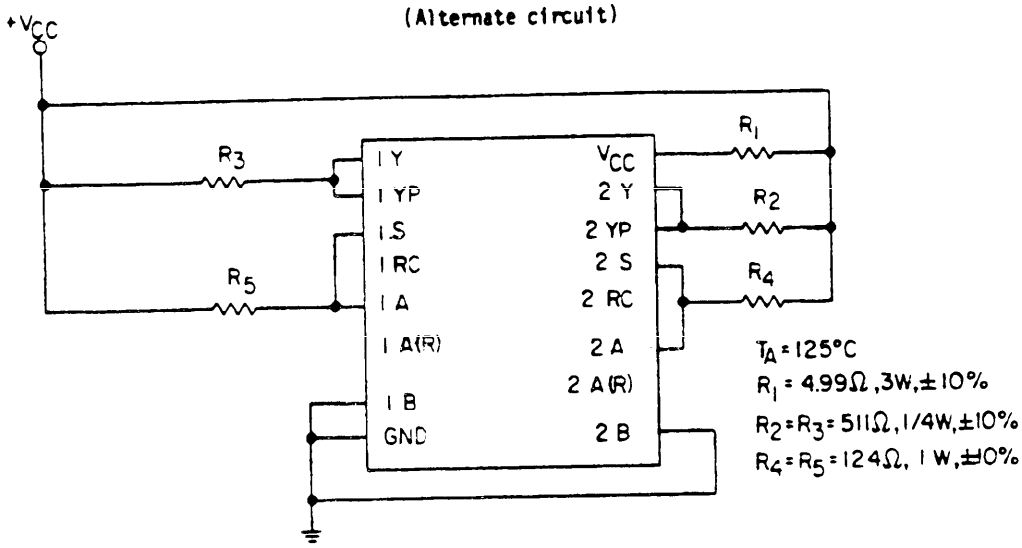
FIGURE 3. Burn-in and life-test circuits - Continued.

NOTES:
1. All resistors are 1/2 watt, ±5 percent.
2. $+V_{CC}$ shall be a minimum of +5.0 V when measured at the $+V_{CC}$ terminal of the device.

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Device type 04

(Alternate circuit)



Device type 05

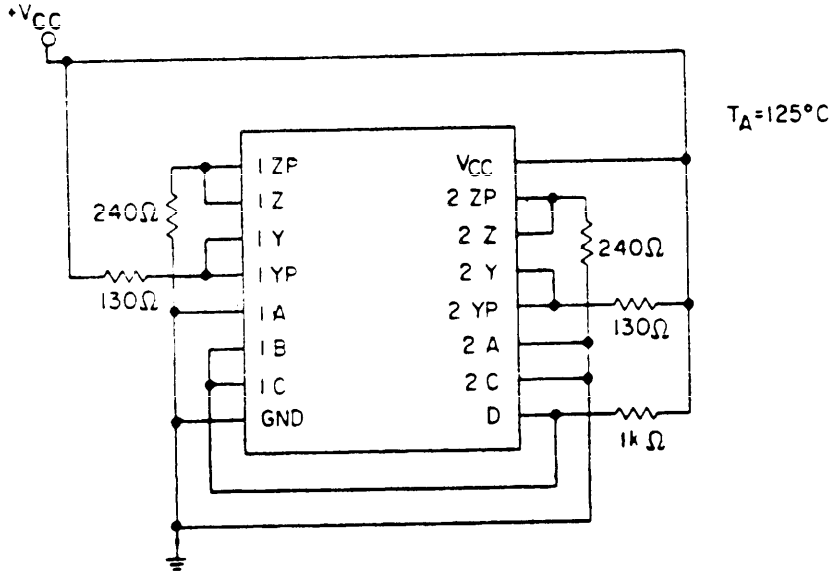
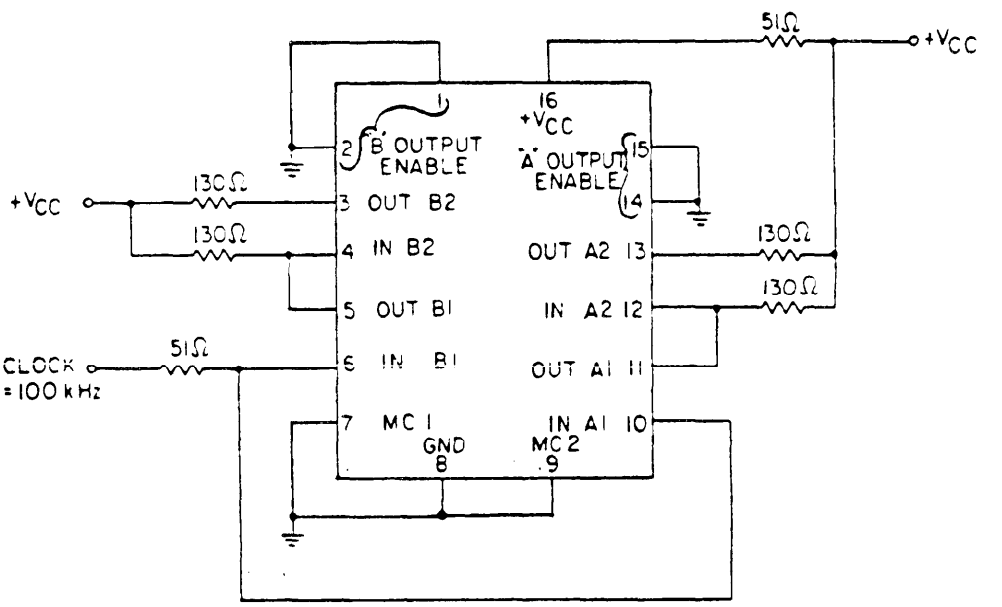


FIGURE 3. Burn-in and life-test circuits - Continued.

NOTES:
1. All resistors are 1/2 watt, ±5 percent.
2. $+V_{CC}$ shall be a minimum of +5.0 V when measured at the $+V_{CC}$ terminal of the device.

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Device types 06 and 07

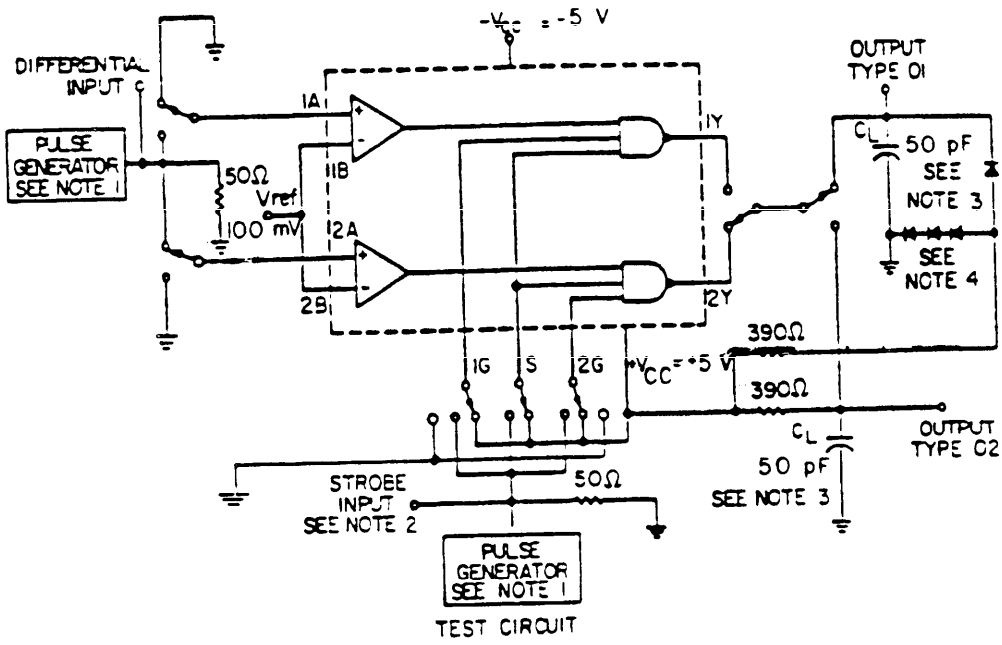


NOTES:
1. All resistors are 1/2 watt, ±5 percent.
2. All clock pulses are typically 0 to 3.0 volts, 50 percent duty cycle.
3. $+V_{CC}$ shall be a minimum of +5.0 V when measured at the $+V_{CC}$ terminal of the device.

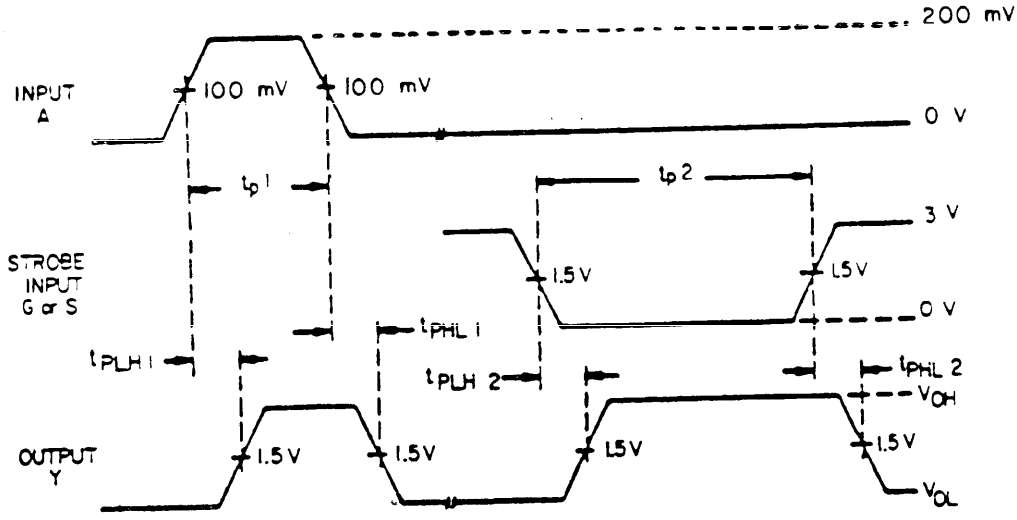
FIGURE 3. Burn-in and life-test circuits - Continued.

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Device types 01 and 02



TEST CIRCUIT



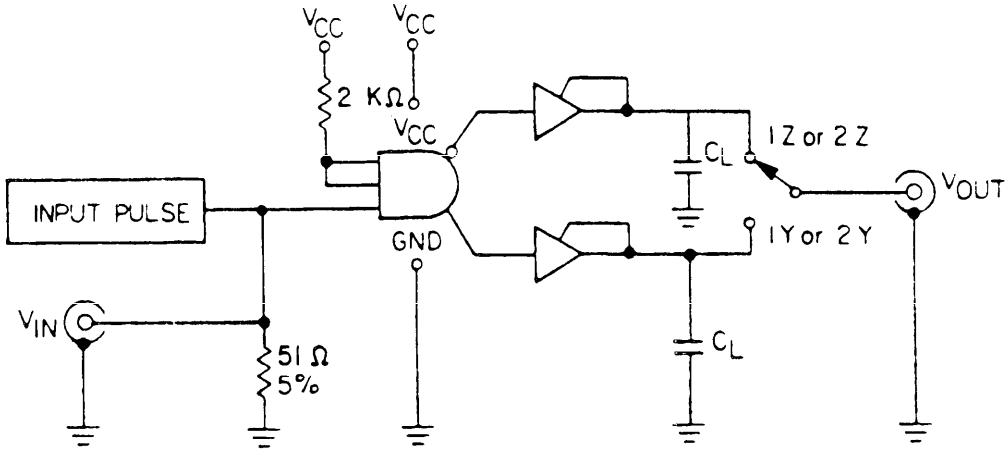
VOLTAGE WAVEFORMS

NOTES:
1. The pulse generators have the following characteristics: Z_{out} = 50Ω, $t_{TLH} = t_{THL} = \leq 5$ ns, t_{p1} = 500 ns, PRR = 1 MHz, t_{p2} = 1 µs, PRR = 500 kHz.
2. All strobe inputs are tested according to table III.
3. C_L includes probe and jig capacitance.
4. All diodes are 1N916, or equivalent.

FIGURE 4. Switching time test circuit.

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Device type 03



NOTES:
1. Input pulse: Frequency = 500 kHz, amplitude = 3 ±.1 V, pulse width = 110 ±10 ns, $t_r = t_f \leq 5$ ns.
2. C_L = 50 pF including probe and jig capacitance.
3. V_{CC} = 5 V.

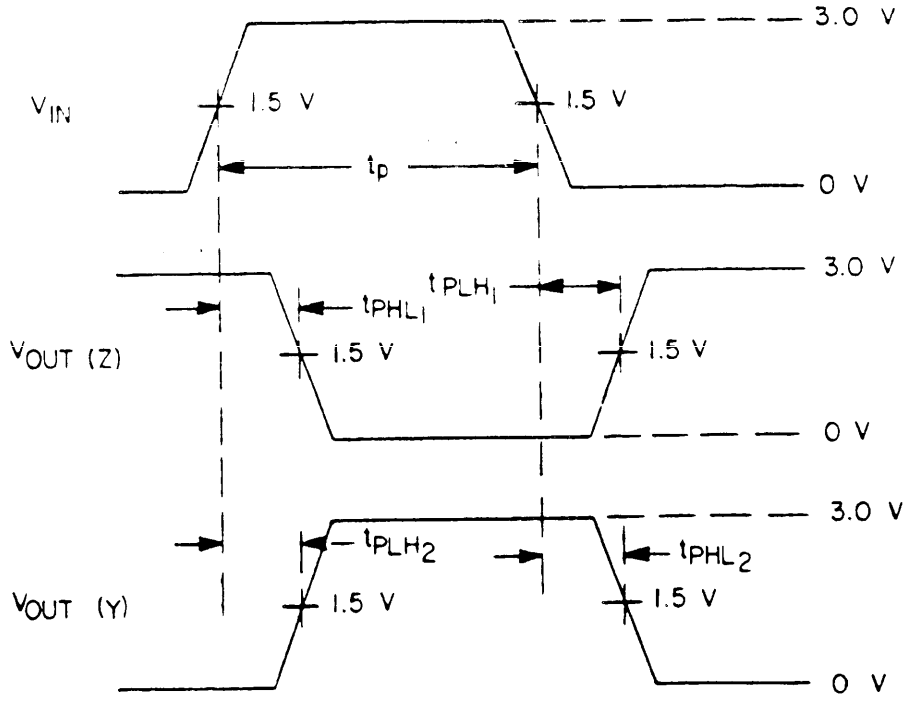
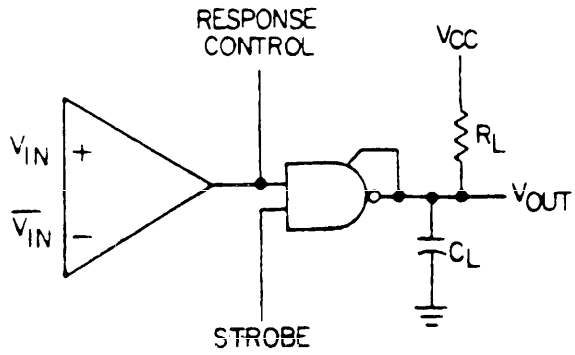
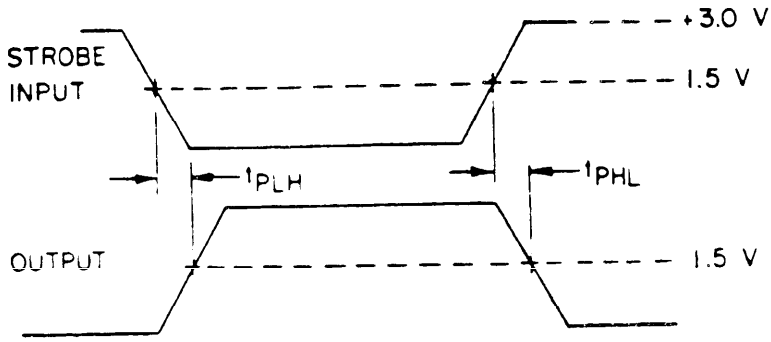


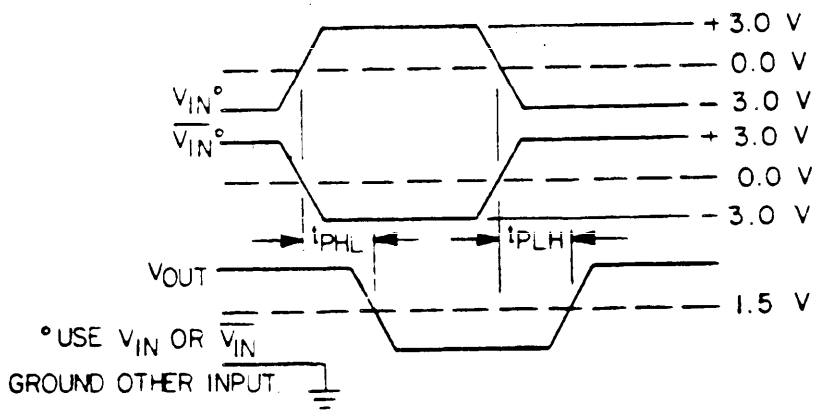
FIGURE 4. Switching time test circuit - Continued.

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Device type 04





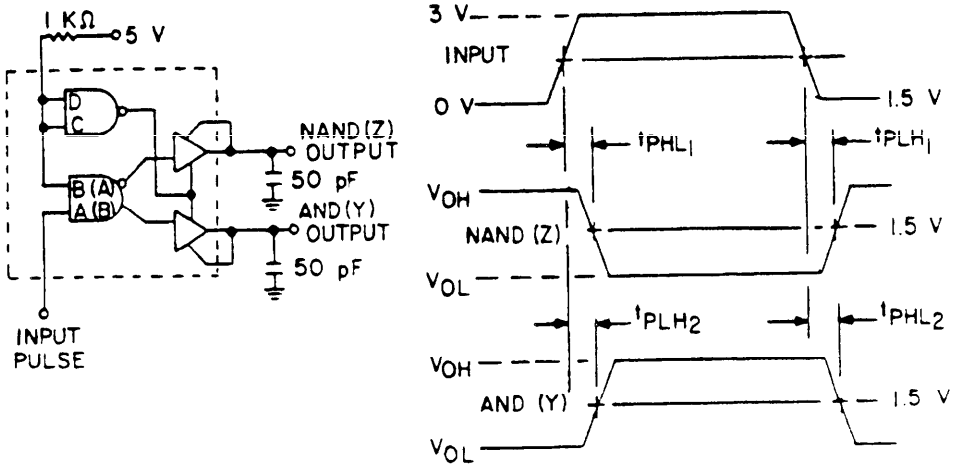


NOTES:
1. For t_{PHL} measurement, R_L = 390Ω.
2. For t_{PLH} measurement, R_L = 3.9 kΩ.
3. Input pulse: Pulse width 100 ns ±10 ns, $t_r = t_f = \leq 5$ ns, PRR = 500 kHz.
4. C_L = 50 pF including probe and jig capacitance.
5. Response control open, maximum socket capacitance = 5 pF.

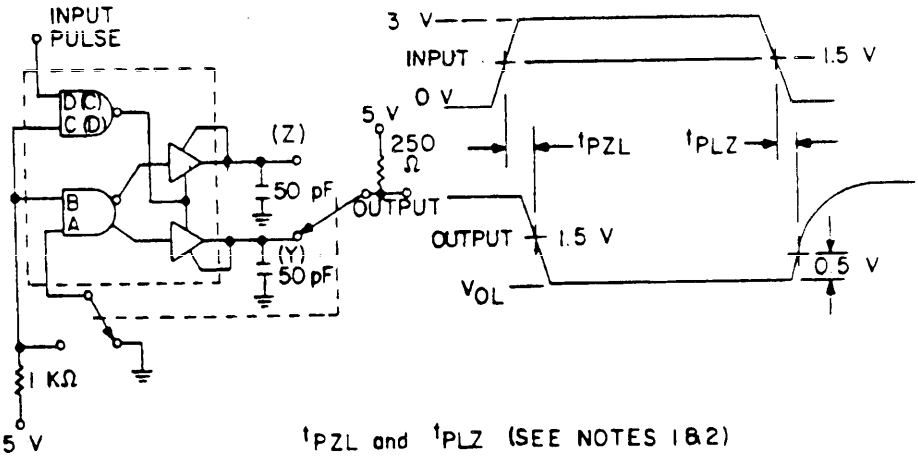
FIGURE 4. Switching time test circuit - Continued.

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Device type 05



t_{PLH} and t_{PHL} (SEE NOTES 1&2)



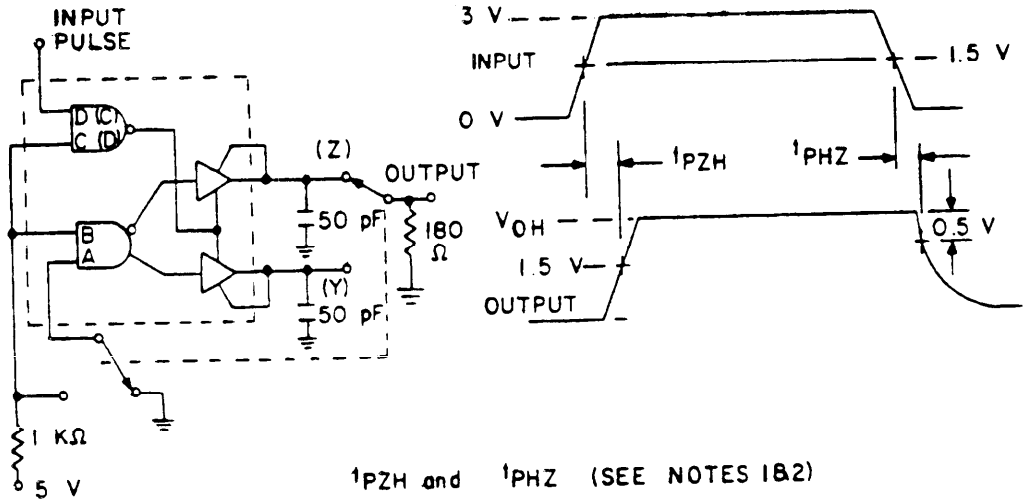
t_{PZL} and t_{PLZ} (SEE NOTES 1&2)

See notes at end of device type 05.

FIGURE 4. Switching time test circuit - Continued.

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Device type 05

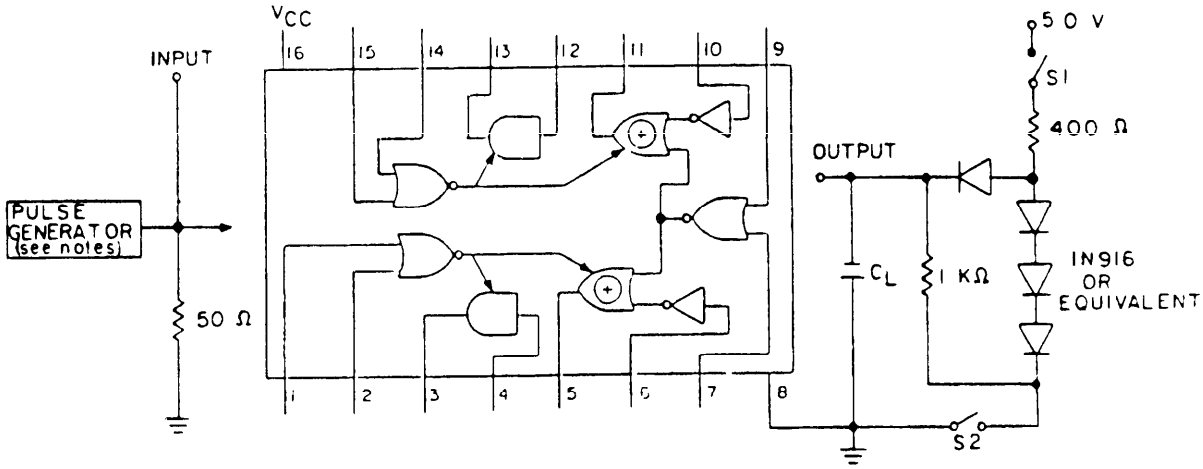


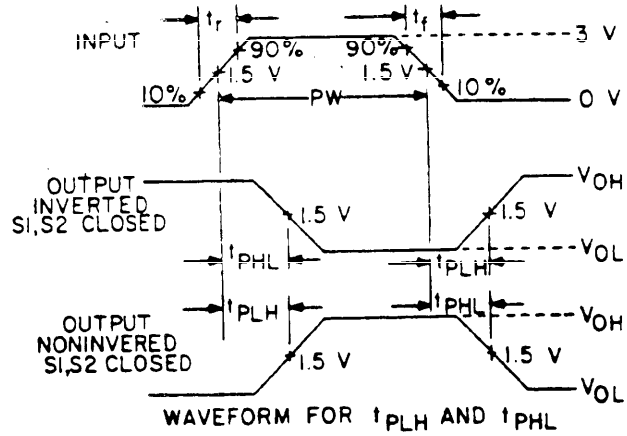
NOTES:
1. The input pulse has the following characteristics: $t_r \leq 5$ ns, $t_f \leq 5$ ns, t_p = 100 ns, PRR = 500 kHz.
2. C_L includes probe and test fixture capacitance.

FIGURE 4. Switching time test circuit - Continued.

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Device types 06 and 07



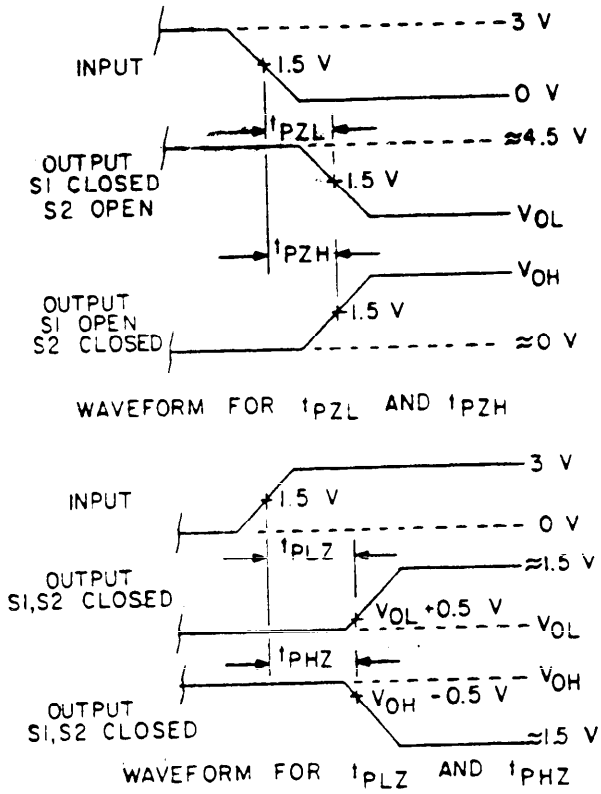


See notes at end of device types 06 and 07

FIGURE 4. Switching time test circuit - Continued.

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Device types 06 and 07



NOTES:
1. C_L = includes probe and jig capacitance = 50 pF, ±10 percent.
2. Input characteristics: $t_r = t_f$ = ≤ 5 ns, PW = 500 µs, PRR ≤ 500 kHz.
3. All resistors are ± 5 percent.

FIGURE 4. Switching time test circuit - Continued.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, D Case C	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
				1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	
			Test no.	1A	1B	NC	1Y	1G	S	GND	2G	2Y	NC	2B	2A	$-V_{CC}$	$+V_{CC}$				
1 T_A = +25°C	V_{OH}	3006	1	1/	+3.0 V		-400 µA	2.0 V	2.0 V	GND	GND			GND	GND	-4.5 V	+4.5 V	1Y	2.4		V
			2	1/	-3.0 V		"	2.0 V	"	"	"			"	"	"	"	"	"		"
			3	2/	+3.0 V		"	0.8 V	"	"	"			"	"	"	"	"	"		"
			4	"	-3.0 V		"	0.8 V	"	"	"			"	"	"	"	"	"		"
			5	"	+3.0 V		"	2.0 V	0.8 V	"	"			"	"	"	"	"	"		"
			6	"	-3.0 V		"	2.0 V	0.8 V	"	"			"	"	"	"	"	"		"
			7	GND	GND			GND	2.0 V	"	2.0 V	-400 µA		+3.0 V	1/	"	"	2Y	"		"
			8	"	"			"	"	"	2.0 V	"		-3.0 V	1/	"	"	"	"		"
			9	"	"			"	"	"	0.8 V	"		+3.0 V	2/	"	"	"	"		"
			10	"	"			"	"	"	0.8 V	"		-3.0 V	"	"	"	"	"		"
			11	"	"			"	0.8 V	"	2.0 V	"		+3.0 V	"	"	"	"	"		"
			12	"	"			"	0.8 V	"	2.0 V	"		-3.0 V	"	"	"	"	"		"
	V_{OL}	3007	13	2/	+3.0 V		16 mA	2.0 V	2.0 V	"	GND			GND	GND	"	"	1Y		0.4	"
			14	2/	-3.0 V		16 mA	2.0 V	"	"	GND			GND	GND	"	"	1Y		"	"
			15	GND	GND			GND	"	"	2.0 V	16 mA		+3.0 V	2/	"	"	2Y		"	"
			16	GND	GND			GND	"	"	2.0 V	16 mA		-3.0 V	2/	"	"	2Y		"	"
	I_{IH1}	3010	17	3/	+3.0 V				GND	"				GND	GND	-5.5 V	+5.5 V	1A		75	µA
			18	3/	-3.0 V				"	"				GND	GND	"	"	1A		"	"
			19	GND	GND				"	"				+3.0 V	3/	"	"	2A		"	"
			20	GND	"				"	"				-3.0 V	3/	"	"	2A		"	"
	I_{IH2}		21	25 mV	"			GND	2.4 V	"	GND			GND	25 mV	"	"	S		MO	"
	I_{IH3}		22	"	"			GND	5.5 V	"	"			"	25 mV	"	"	S		2	mA
	I_{IH4}		23	"	"			2.4 V	GND	"	"			"	GND	"	"	1G		40	µA
	I_{IH4}		24	GND	"			GND	"	"	2.4 V			"	25 mV	"	"	2G		40	"
	I_{IH5}		25	25 mV	"			5.5 V	"	"	GND			"	GND	"	"	1G		100	"
	I_{IH5}		26	GND	"			GND	"	"	5.5 V			"	25 mV	"	"	2G		100	"
	I_{IL1}	3009	27	4/	+3.0 V					"				GND	GND	"	"	1A		-10	"
			28	4/	-3.0 V					"				GND	GND	"	"	1A		"	"
			29	GND	GND					"				+3.0 V	4/	"	"	2A		"	"
			30	GND	"					"				-3.0 V	4/	"	"	2A		"	"
	I_{IL2}		31	-25 mV	"			0.4 V	4.5 V	"	GND			GND	GND	"	"	1G	-0.7	-1.6	mA
	I_{IL2}		32	GND	"			GND	4.5 V	"	0.4 V			"	-25 mV	"	"	2G	-0.7	-1.6	"
	I_{IL3}		33	-25 mV	"			4.5 V	0.4 V	"	4.5 V			"	-25 mV	"	"	S	-1.4	-3.2	"
	I_{OS}	3011	34	25 mV	"		GND	GND	GND	"	GND			"	+25 mV	"	"	1Y	-18	-70	"
	I_{OS}		35	"	"			GND	GND	"	GND	GND		"	"	"	"	2Y	-18	-70	"
	I_{CCH+}	3005	36	"	"			5.0 V	5.0 V	"	5.0 V			"	"	"	"	$+V_{CC}$		+30	"
	I_{CCH-}	3005	37	"	"			5.0 V	5.0 V	"	5.0 V			"	"	"	"	$-V_{CC}$		-15	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_A = +125°C.																				
3	Same tests, terminal conditions, and limits as subgroup 1, except T_A = -55°C.																				

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,D Case C	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits Min	Limits Max	Unit
			Test no.	1A	1B	NC	1Y	1G	S	GND	2G	2Y	NC	2B	2A	$-V_{CC}$	$+V_{CC}$				
9 T_A = +25°C	t_{PLH1}	3003 (Fig. 4)	38	IN	100 mV		OUT	5.0 V	5.0 V	GND	GND			100 mV	GND	-5.0 V	+5.0 V	1A to 1Y		25	ns
	t_{PLH1}		39	GND	"			GND	"	"	5.0 V	OUT		"	IN	"	"	2A to 2Y		"	"
	t_{PHL1}		40	IN	"		OUT	5.0 V	"	"	GND			"	GND	"	"	1A to 1Y		"	"
	t_{PHL1}		41	GND	"			GND	"	"	5.0 V	OUT		"	IN	"	"	2A to 2Y		"	"
	t_{PLH2}		42	"	"		OUT	IN	"	"	GND			"	GND	"	"	1G to 1Y		20	"
	t_{PLH2}		43	"	"			GND	"	"	IN	OUT		"	"	"	"	2G to 2Y		"	"
	t_{PLH2}		44	"	"		OUT	5.0 V	IN	"	GND			"	"	"	"	S to 1Y		"	"
	t_{PLH2}		45	"	"			GND	IN	"	5.0 V	OUT		"	"	"	"	S to 2Y		"	"
	t_{PHL2}		46	"	"		OUT	IN	5.0 V	"	GND			"	"	"	"	1G to 1Y		15	"
	t_{PHL2}		47	"	"			GND	5.0 V	"	IN	OUT		"	"	"	"	2G to 2Y		"	"
	t_{PHL2}		48	"	"		OUT	5.0 V	IN	"	GND			"	"	"	"	S to 1Y		"	"
	t_{PHL2}		49	"	"			GND	IN	"	5.0 V	OUT		"	"	"	"	S to 2Y		"	"
10 T_A = +125°C	t_{PLH1}		50	IN	"		OUT	5.0 V	5.0 V	"	GND			"	GND	"	"	1A to 1Y		40	"
	t_{PLH1}		51	GND	"			GND	"	"	5.0 V	OUT		"	IN	"	"	2A to 2Y		40	"
	t_{PHL1}		52	IN	"		OUT	5.0 V	"	"	GND			"	GND	"	"	1A to 1Y		35	"
	t_{PHL1}		53	GND	"			GND	"	"	5.0 V	OUT		"	IN	"	"	2A to 2Y		35	"
	t_{PLH2}		54	"	"		OUT	IN	"	"	GND			"	GND	"	"	1G to 1Y		30	"
	t_{PLH2}		55	"	"			GND	"	"	IN	OUT		"	"	"	"	2G to 2Y		"	"
	t_{PLH2}		56	"	"		OUT	5.0 V	IN	"	GND			"	"	"	"	S to 1Y		"	"
	t_{PLH2}		57	"	"			GND	IN	"	5.0 V	OUT		"	"	"	"	S to 2Y		"	"
	t_{PHL2}		58	"	"		OUT	IN	5.0 V	"	GND			"	"	"	"	1G to 1Y		25	"
	t_{PHL2}		59	"	"			GND	5.0 V	"	IN	OUT		"	"	"	"	2G to 2Y		"	"
	t_{PHL2}		60	"	"		OUT	5.0 V	IN	"	GND			"	"	"	"	S to 1Y		"	"
	t_{PHL2}		61	"	"			GND	IN	"	5.0 V	OUT		"	"	"	"	S to 2Y		"	"
11	Same tests, terminals conditions, and limits as for subgroup 10, except T_A = -55°C																				

1/ +25 mV applied from input 1A to input 1B or from input 2A to input 2B, as applicable.
2/ -25 mV applied from input 1A to input 1B or from input 2A to input 2B, as applicable.
3/ 0.5 V applied from input 1A to input 1B or from input 2A to input 2B, as applicable.
4/ -2.0 V applied from positive input to negative input.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,D / Case C	1 / 1	2 / 1A	3 / 1B	4 / NC	5 / 1Y	6 / 1G	7 / S	8 / GND	9 / 2G	10 / 2Y	11 / NC	12 / 2B	13 / 2A	14 / -V_{CC}	+V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
			Test no.																			
1 T_A = +25°C	I_{OH}		1	1/	+3.0 V		5.5 V	2.0 V	2.0 V	GND	GND			GND	GND	-4.5 V	+4.5 V	1Y		250	µA
			2	1/	-3.0 V		"	2.0 V	"	"	"			"	"	"	"	"		"	"
			3	2/	+3.0 V		"	0.8 V	"	"	"			"	"	"	"	"		"	"
			4	"	-3.0 V		"	0.8 V	"	"	"			"	"	"	"	"		"	"
			5	"	+3.0 V		"	2.0 V	0.8 V	"	"			"	"	"	"	"		"	"
			6	"	-3.0 V		"	2.0 V	0.8 V	"	"			"	"	"	"	"		"	"
			7	GND	GND		"	GND	2.0 V	"	2.0 V	5.5 V		+3.0 V	1/	"	"	2Y		"	"
			8	"	"		"	"	"	"	2.0 V	"		-3.0 V	1/	"	"	"		"	"
			9	"	"		"	"	"	"	0.8 V	"		+3.0 V	2/	"	"	"		"	"
			10	"	"		"	"	"	"	0.8 V	"		-3.0 V	"	"	"	"		"	"
			11	"	"		"	"	0.8 V	"	2.0 V	"		+3.0 V	"	"	"	"		"	"
			12	"	"		"	"	0.8 V	"	2.0 V	"		-3.0 V	"	"	"	"		"	"
	V_{OL}	3007	13	2/	+3.0 V		16 mA	2.0 V	2.0 V	"	GND			GND	GND	"	"	1Y		0.4	V
			14	2/	-3.0 V		16 mA	2.0 V	"	"	GND			GND	GND	"	"	1Y		"	"
			15	GND	GND			GND	"	"	2.0 V	16 mA		+3.0 V	2/	"	"	2Y		"	"
			16	GND	GND			GND	"	"	2.0 V	16 mA		-3.0 V	2/	"	"	2Y		"	"
	I_{IH1}	3010	17	3/	+3.0 V				GND	"				GND	GND	-5.5 V	+5.5 V	1A		75	µA
			18	3/	-3.0 V				"	"				GND	GND	"	"	1A		"	"
			19	GND	GND				"	"				+3.0 V	3/	"	"	2A		"	"
			20	GND	"				"	"				-3.0 V	3/	"	"	2A		"	"
	I_{IH2}		21	25 mV	"			GND	2.4 V	"	GND			GND	25 mV	"	"	S		80	"
	I_{IH3}		22	"	"			GND	5.5 V	"	"			"	25 mV	"	"	S		2	mA
	I_{IH4}		23	"	"			2.4 V	GND	"	"			"	GND	"	"	1G		40	µA
	I_{IH4}		24	GND	"			GND	"	"	2.4 V			"	25 mV	"	"	2G		40	"
	I_{IH5}		25	25 mV	"			5.5 V	"	"	GND			"	GND	"	"	1G		100	"
	I_{IH5}		26	GND	"			GND	"	"	5.5 V			"	25 mV	"	"	2G		100	"
	I_{IL1}	3009	27	4/	+3.0 V					"				GND	GND	"	"	1A		-10	"
			28	4/	-3.0 V					"				GND	GND	"	"	1A		"	"
			29	GND	GND					"				+3.0 V	4/	"	"	2A		"	"
			30	GND	"					"				-3.0 V	4/	"	"	2A		"	"
	I_{IL2}		31	-25 mV	"			0.4 V	4.5 V	"	GND			GND	GND	"	"	1G	-0.7	-1.6	mA
	I_{IL2}		32	GND	"			GND	4.5 V	"	0.4 V			"	-25 mV	"	"	2G	-0.7	-1.6	"
	I_{IL3}		33	-25 mV	"			4.5 V	0.4 V	"	4.5 V			"	-25 mV	"	"	S	-1.4	-3.2	"
	I_{CCH+}	3005	34	+25 mV	"			5.0 V	5.0 V	"	5.0 V			"	+25 mV	"	"	+V_{CC}		+30	"
	I_{CCH-}	3005	35	+25 mV	"			5.0 V	5.0 V	"	5.0 V			"	+25 mV	"	"	-V_{CC}		-15	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_A = +125°C.																					
3	Same tests, terminal conditions, and limits as subgroup 1, except T_A = -55°C.																					

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,D / Case C / Test no.	1 / 1A	2 / 1B	3 / NC	4 / 1Y	5 / 1G	6 / S	7 / GND	8 / 2G	9 / 2Y	10 / NC	11 / 2B	12 / 2A	13 / $-V_{CC}$	14 / $+V_{CC}$	Measured terminal	Limits Min	Limits Max	Unit
9 T_A = +25°C	t_{PLH1}	3003 (Fig. 4)	36	IN	100 mV		OUT	5.0 V	5.0 V	GND	GND			100 mV	GND	-5.0 V	+5.0 V	1A to 1Y		25	ns
	t_{PLH1}		37	GND	"			GND	"	"	5.0 V	OUT		"	IN	"	"	2A to 2Y		"	"
	t_{PHL1}		38	IN	"		OUT	5.0 V	"	"	GND			"	GND	"	"	1A to 1Y		"	"
	t_{PHL1}		39	GND	"			GND	"	"	5.0 V	OUT		"	IN	"	"	2A to 2Y		"	"
	t_{PLH2}		40	"	"		OUT	IN	"	"	GND			"	GND	"	"	1G to 1Y		"	"
	t_{PLH2}		41	"	"			GND	"	"	IN	OUT		"	"	"	"	2G to 2Y		"	"
	t_{PLH2}		42	"	"		OUT	5.0 V	IN	"	GND			"	"	"	"	S to 1Y		"	"
	t_{PHL2}		43	"	"			GND	IN	"	5.0 V	OUT		"	"	"	"	S to 2Y		"	"
	t_{PHL2}		44	"	"		OUT	IN	5.0 V	"	GND			"	"	"	"	1G to 1Y		20	"
	t_{PHL2}		45	"	"			GND	5.0 V	"	IN	OUT		"	"	"	"	2G to 2Y		"	"
	t_{PHL2}		46	"	"		OUT	5.0 V	IN	"	GND			"	"	"	"	S to 1Y		"	"
	t_{PHL2}		47	"	"			GND	IN	"	5.0 V	OUT		"	"	"	"	S to 2Y		"	"
10 T_A = +125°C	t_{PLH1}		48	IN	"		OUT	5.0 V	5.0 V	"	GND			"	GND	"	"	1A to 1Y		45	"
	t_{PLH1}		49	GND	"			GND	"	"	5.0 V	OUT		"	IN	"	"	2A to 2Y		45	"
	t_{PHL1}		50	IN	"		OUT	5.0 V	"	"	GND			"	GND	"	"	1A to 1Y		35	"
	t_{PHL1}		51	GND	"			GND	"	"	5.0 V	OUT		"	IN	"	"	2A to 2Y		35	"
	t_{PLH2}		52	"	"		OUT	IN	"	"	GND			"	GND	"	"	1G to 1Y		40	"
	t_{PLH2}		53	"	"			GND	"	"	IN	OUT		"	"	"	"	2G to 2Y		"	"
	t_{PLH2}		54	"	"		OUT	5.0 V	IN	"	GND			"	"	"	"	S to 1Y		"	"
	t_{PHL2}		55	"	"			GND	IN	"	5.0 V	OUT		"	"	"	"	S to 2Y		"	"
	t_{PHL2}		56	"	"		OUT	IN	5.0 V	"	GND			"	"	"	"	1G to 1Y		25	"
	t_{PHL2}		57	"	"			GND	5.0 V	"	IN	OUT		"	"	"	"	2G to 2Y		"	"
	t_{PHL2}		58	"	"		OUT	5.0 V	IN	"	GND			"	"	"	"	S to 1Y		"	"
	t_{PHL2}		59	"	"			GND	IN	"	5.0 V	OUT		"	"	"	"	S to 2Y		"	"
11	Same tests, terminals conditions, and limits as for subgroup 10, except T_A = -55°C																				

1/ +25 mV applied from input 1A to input 1B or from input 2A to input 2B, as applicable.
2/ -25 mV applied from input 1A to input 1B or from input 2A to input 2B, as applicable.
3/ 0.5 V applied from input 1A to input 1B or from input 2A to input 2B, as applicable.
4/ -2.0 V applied from positive input to negative input.

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Test no.	1ZP	1Z	1Y	1YP	1A	1B	1C	GND	2A	2B	2C	2YP	2Y	2Z	2ZP	V_{CC}		Min	Max	
1 T_A = +25°C	V_{OL}	3007	1	1Z	40 mA			2.0 V	2.0 V	2.0 V	GND								4.5 V	1Z		0.4	V
			2								"	2.0 V	2.0 V	2.0 V			40 mA	2Z	"	2Z		"	"
			3			40 mA	1Y	0.8 V	2.0 V	2.0 V	"								"	1Y		"	"
			4			"	"	2.0 V	0.8 V	2.0 V	"								"	"		"	"
			5			"	"	2.0 V	2.0 V	0.8 V	"								"			"	"
			6								"	0.8 V	2.0 V	2.0 V	2Y	40 mA			"	2Y		"	"
			7								"	2.0 V	0.8 V	2.0 V	"	"			"	"		"	"
			8								"	2.0 V	2.0 V	0.8 V	"	"			"	"		"	"
	V_{OH1}	3006	9			1YP	-10 mA	2.0 V	2.0 V	2.0 V	"	GND	GND	GND					"	1YP	2.4		"
			10					GND	GND	GND	"	2.0 V	2.0 V	2.0 V	-10 mA	2YP			"	2YP	"		"
			11	-10 mA	1ZP			0.8 V	2.0 V	2.0 V	"	GND	GND	GND					"	1ZP	"		"
			12	"	"			2.0 V	0.8 V	2.0 V	"	"	"	"					"	"	"		"
			13	"	"			2.0 V	2.0 V	0.8 V	"	"	"	"					"	"	"		"
			14					GND	GND	GND	"	0.8 V	2.0 V	2.0 V			2ZP	-10 mA	"	2ZP	"		"
			15					"	"	"	"	2.0 V	0.8 V	2.0 V			"	"	"	"	"		"
			16					"	"	"	"	2.0 V	2.0 V	0.8 V			"	"	"	"	"		"
	V_{OH2}	3006	17			1YP	-20 mA	2.0 V	2.0 V	2.0 V	"	GND	GND	GND					"	1YP	"		"
			18					GND	GND	GND	"	2.0 V	2.0 V	2.0 V	-20 mA	2YP			"	2YP	"		"
			19	-20 mA	1ZP			0.8 V	2.0 V	2.0 V	"	GND	GND	GND					"	1ZP	"		"
			20	"	"			2.0 V	0.8 V	2.0 V	"	"	"	"					"	"	"		"
			21	"	"			2.0 V	2.0 V	0.8 V	"	"	"	"					"	"	"		"
			22					GND	GND	GND	"	0.8 V	2.0 V	2.0 V			2ZP	-20 mA	"	2ZP	"		"
			23					"	"	"	"	2.0 V	0.8 V	2.0 V			"	"	"	"	"		"
			24					"	"	"	"	2.0 V	2.0 V	0.8 V			"	"	"	"	"		"
	I_{OS}	3011	25	GND	1ZP			0.8 V	0.8 V	0.8 V	"								5.5 V	1ZP	-40	-130	mA
			26								"	0.8 V	0.8 V	0.8 V			2ZP	GND	"	2ZP	"	"	"
			27			1YP	GND	2.0 V	2.0 V	2.0 V	"								"	1YP	"	"	"
			28								"	2.0 V	2.0 V	2.0 V	GND	2YP			"	2YP	"	"	"
	I_{CC}	3005	29					GND	GND	GND	"	GND	GND	GND					"	V_{CC}		50	"
	I_{CEX}		30		12 V			0.8 V	0.8 V	0.8 V	"								"	1Z		100	μA
			31								"	0.8 V	0.8 V	0.8 V			12 V		"	2Z		"	"
			32			12 V		2.0 V	2.0 V	2.0 V	"								"	1Y		"	"
			33								"	2.0 V	2.0 V	2.0 V		12 V			"	2Y		"	"
	I_{IL}	3009	34					0.4 V	5.5 V	5.5 V	"	5.5 V	5.5 V	5.5 V					"	1A	-0.7	-1.6	mA
			35					5.5 V	0.4 V	5.5 V	"	"	"	"					"	1B	"	"	"
			36					"	5.5 V	0.4 V	"	"	"	"					"	1C	"	"	"
			37					"	"	5.5 V	"	"	"	"					"	2A	"	"	"
			38					"	"	"	"	0.4 V	"	"					"	2B	"	"	"
			39					"	"	"	"	5.5 V	0.4 V	0.4 V					"	2C	"	"	"
	I_{IH1}	3010	40					2.4 V	GND	GND	"	GND	GND	GND					"	1A		40	μA
			41					GND	2.4 V	GND	"	"	"	"					"	1B		"	"
			42					"	GND	2.4 V	"	"	"	"					"	1C		"	"
			43					"	"	GND	"	2.4 V	"	"					"	2A		"	"
			44					"	"	"	"	GND	2.4 V	"					"	2B		"	"
			45					"	"	"	"	GND	GND	2.4 V					"	2C		"	"
	I_{IH2}	3010	46					5.5 V	GND	GND	"	GND	GND	GND					"	1A		100	"
			47					GND	5.5 V	GND	"	"	"	"					"	1B		"	"
			48					"	GND	5.5 V	"	"	"	"					"	1C		"	"
			49					"	"	GND	"	5.5 V	"	"					"	2A		"	"
			50					"	"	"	"	GND	5.5 V	"					"	2B		"	"
			51					"	"	"	"	GND	GND	5.5 V					"	2C		"	"

See footnote at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Test no.	1ZP	1Z	1Y	1YP	1A	1B	1C	GND	2A	2B	2C	2YP	2Y	2Z	2ZP	V_{CC}		Min	Max	
1 T_A = +25°C	$\|8V_{IN}\|$		52 53 54 55 56 57					1 mA	1 mA	1 mA	GND " " " " "	1 mA	1 mA	1 mA					5.5 V " " " " "	1A 1B 1C 2A 2B 2C	5.5 " " " " "		V " " " " "
	V_{IC}		58 59 60 61 62 63					-12 mA	-12 mA	-12 mA	" " " " " "	-12 mA	-12 mA	-12 mA					" " " " " "	1A 1B 1C 2A 2B 2C		-1.5 " " " " "	" " " " " "
	V_{OC}		64 65 66 67		-40 mA	-40 mA		GND	GND	GND	" " " "	GND	GND	GND		-40 mA	-40 mA		" " " "	1Z 2Z 1Y 2Y		" " " "	" " " "
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_A = +125°C and V_{IC} and V_{OC} tests are omitted and maximum limits for I_{CEX} is 200 µA.																						
3	Same tests, terminal conditions, and limits as for subgroup 2, except T_A = -55°C and minimum limits for V_{OH2} is 2.2 V, and maximum limits for I_{OS} is -155 mA.																						
9 T_A = +25°C	t_{PHL1} t_{PLH1} t_{PHL1} t_{PLH1} t_{PHL1} t_{PLH1} t_{PHL1} t_{PLH1} t_{PHL1} t_{PLH1} t_{PHL1} t_{PLH1}	3003 (Fig. 4)	68 69 70 71 72 73 74 75 76 77 78 79	1Z " " " " "	OUT " " " " "			IN IN 1/ " " "	1/ 1/ IN IN 1/ 1/	1/ " " IN IN	GND " " " " " " " " " " " "	IN IN 1/ " " "	1/ 1/ IN IN 1/ 1/	1/ " " IN IN			OUT " " " " "	2Z " " " " "	5.0 V " " " " " " " " " " " "	1Z " " " 2Z " " " " " " "		25 " " " " " " " " " " "	ns " " " " " " " " " " "
	t_{PHL2} t_{PLH2} t_{PHL2} t_{PLH2} t_{PHL2} t_{PLH2} t_{PHL2} t_{PLH2} t_{PHL2} t_{PLH2} t_{PHL2} t_{PLH2}	3003 (Fig. 4)	80 81 82 83 84 85 86 87 88 89 90 91			OUT " " " " "	1Y " " " " "	IN IN 1/ " " "	1/ 1/ IN IN 1/ 1/	1/ " " IN IN	" " " " " " " " " " " "	IN IN 1/ " " "	1/ 1/ IN IN 1/ 1/	1/ " " IN IN	2Y " " " " "	OUT " " " " "			" " " " " " " " " " " "	1Y " " " 2Y " " " " " " "		" " " " " " " " " " " "	" " " " " " " " " " " "
10	Same tests, terminal conditions, and limits as for subgroup 9, except T_A = +125°C and the limits for t_{PLH1} and t_{PHL2} are 12 ns maximum, and for t_{PHL1} and t_{PLH2} are 30 ns maximum.																						
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_A = -55°C																						

1/ Indicates pins are connected together and to a 2 kΩ resistor to 5.0 V.

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TABLE III. Group A inspection for device type 04.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case 2 / Cases E and F	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Limits		Unit
			Test no.	1Y	1YP	1S	1RC	1A	1A(R)	1B	GND	2B	2A(R)	2A	2RC	2S	3YP	2Y	V_{CC}		Min	Max	
1 T_A = +25°C	V_{OH}	3006	1	1YP	-5 mA			GND		0.5 V	GND	0.5 V		GND			-5 mA	2YP	4.5 V	1YP	2.4		V
			2					GND		0.5 V	"	0.5 V		GND					"	2YP	"		"
			3	1YP	-5 mA	0.5 V		0.5 V		GND	"	GND		0.5 V					"	1YP	"		"
			4					0.5 V		GND	"	GND		0.5 V		0.5 V	-5 mA	2YP	"	2YP	"		"
			5	1YP	-5 mA			15 V		17 V	"								"	1YP	"		"
			6								"	17 V		15 V			-5 mA	2YP	"	2YP	"		"
			7	1YP	-5 mA			-17 V		-15 V	"								"	1YP	"		"
			8								"	-15 V		-17 V			-5 mA	2YP	"	2YP	"		"
	V_{OL}	3007	9	16 mA	1Y			0.5 V		GND	"	GND		0.5 V					"	1Y		0.4	"
			10					0.5 V		GND	"			0.5 V			3Y	16 mA	"	3Y		"	"
			11	16 mA	1Y			17 V		15 V	"	GND							"	1Y		"	"
			12								"			17 V			3Y	16 mA	"	3Y		"	"
			13	16 mA	1Y			-15 V		-17 V	"	15 V							"	1Y		"	"
			14								"			15 V			3Y	16 mA	"	3Y		"	"
												-17 V		-15 V									
	I_{CC}	3005	15					0.5 V		GND	"	GND		0.5 V					5.5 V	V_{CC}		50	mA
	ΔV_{IN}	3006	16			1 mA		"		"	"	"		"					"	1S	5.5		V
	ΔV_{IN}	3006	17					"		"	"	"		"		1 mA			"	2S	5.5		"
	V_{IC}		18			-12 mA		"		"	"	"		"					4.5 V	1S		-1.5	"
	V_{IC}		19					"		"	"	"		"		-12 mA			4.5 V	2S		-1.5	"
	I_{IL1}	3009	20					0.4 V		5.5 V	"								5.5 V	1A	-0.2	-0.9	mA
			21					5.5 V		0.4 V	"								"	1B	"	"	"
			22								"	0.4 V		5.5 V					"	2B	"	"	"
			23								"	5.5 V		0.4 V					"	2A	"	"	"
	I_{IL2}	3009	24			0.4 V		0.5 V		GND	"								"	1S	-1.0	-2.4	"
	I_{IL2}		25								"	GND		0.5 V		0.4 V			"	2S	-1.0	-2.4	"
	I_{IL3}	3009	26				0.4 V	0.5 V		GND	"								"	1RC	-1.2	-4.1	"
	I_{IL3}		27								"	GND		0.5 V	0.4 V				"	1RC	-1.2	-4.1	"
	I_{IH}	3010	28			4.5 V		0.5 V		GND	"								"	1S		5.0	uA
	I_{IH}	3010	29								"	GND		-0.5 V		4.5 V			"	2S		5.0	"
	I_{CEX}		30	12 V				GND		4.5 V	"								4.5 V	1Y		100	"
	I_{CEX}		31								"	4.5 V		GND				12 V	4.5 V	2Y		100	"
	I_{OS}	3011	32	GND	GND			GND		0.5 V	"								5.5 V	1YP	-15	-80	mA
	I_{OS}	3011	33								"	0.5 V		GND			GND	GND	5.5 V	2YP	-15	-80	mA
	R_{IN}		34					GND	1.0 V		"		1.0 V	GND					5.0 V	1A(R)	77	167	Ω
	R_{IN}		35								"								5.0 V	2A(R)	77	167	Ω
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_A = +125°C, V_{IC} and R_{IN} tests omitted, and maximum limits for I_{CEX} is 200 uA.																						
3	Same tests, terminal conditions, and limits as for subgroup 2, except T_A = -55°C, V_{IC} and R_{IN} tests omitted.																						

See footnote at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Limits		Unit
			Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Min	Max	
			Test no.	1Y	1YP	1S	1RC	1A	1A(R)	1B	GND	2B	2A(R)	2A	2RC	2S	3YP	2Y	V_{CC}				
9 T_A = +25°C 1/	t_{PLH1}	3003 (Fig. 4)	36	OUT	1Y	4.5 V		IN		GND	GND								5.0 V	1Y		50	ns
	t_{PHL1}		37	"	"	"		IN		GND	"								"	"		"	
	t_{PLH1}		38	"	"	"		GND		IN	"								"	"		"	
	t_{PHL1}		39	"	"	"		GND		IN	"	GND		IN		4.5 V	2Y	OUT	"	2Y		"	
	t_{PLH1}		40								"	GND		IN		"	"	"	"	"		"	
	t_{PHL1}		41								"	IN		GND		"	"	"	"	"		"	
	t_{PLH1}		42								"	IN		GND		"			"	1Y		15	
	t_{PHL1}		43								"					"			"	1Y		"	
	t_{PLH2}		44	OUT	1Y	IN		4.5 V		GND	"					"			"	2Y		"	
	t_{PHL2}		45	OUT	1Y	IN		4.5 V		GND	"					IN	2Y	OUT	"	2Y		"	
	t_{PLH2}		46								"	GND		4.5 Y		IN	2Y	OUT	"			"	
	t_{PHL2}		47								"	GND		4.5 Y		IN	2Y	OUT	"			"	
10 T_A = +125°C 1/	t_{PLH1}	3003 (Fig. 4)	48	OUT	1Y	4.5 V		IN		GND	"								"	1Y		60	"
	t_{PHL1}		49	"	"	"		IN		GND	"								"	"		"	
	t_{PLH1}		50	"	"	"		GND		IN	"								"	"		"	
	t_{PHL1}		51	"	"	"		GND		IN	"	GND		IN		4.5 V	2Y	OUT	"	2Y		"	
	t_{PLH1}		52								"	GND		IN		"	"	"	"	"		"	
	t_{PHL1}		53								"	IN		GND		"	"	"	"	"		"	
	t_{PLH1}		54								"	IN		GND		"			"	1Y		25	
	t_{PHL1}		55								"					"			"	1Y		"	
	t_{PLH2}		56	OUT	1Y	IN		4.5 V		GND	"					IN	2Y	OUT	"	2Y		"	
	t_{PHL2}		57	OUT	1Y	IN		4.5 V		GND	"					IN	2Y	OUT	"	2Y		"	
	t_{PLH2}		58								"	GND		4.5 Y									
	t_{PHL2}		59								"	GND		4.5 Y									
11 1/	Same tests, terminal conditions, and limits as for subgroup 10, except T_A = -55°C																						

1/ For subgroups 9, 10, and 11, the unused output terminals may be open or loaded to facilitate equipment setup and minimize switching of loads as long as there is no effect on the measured parameter.

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TABLE III. Group A inspection for device type 05.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits		Unit
			Test no.	1ZP	1Z	1Y	1YP	1A	1B	1C	GND	D	2C	2A	2YP	2Y	2Z	2ZP	V_{CC}		Min	Max	
T_A = 1 +25°C	V_{OH1}	3006	1	-10 mA	1ZP			0.8 V	2.0 V	2.0 V	GND	2.0 V	2.0 V	2.0 V					4.5 V	1ZP	2.4		V
			2	-10 mA	1ZP			2.0 V	0.8 V	"	"	"	"	"					"	1ZP	"		"
			3					"	2.0 V	"	"	"	"	0.8 V			2ZP	-10 mA	"	2ZP	"		"
			4			1YP	-10 mA	"	"	"	"	"	"	2.0 V			"	"	"	1YP	"		"
			5					"	"	"	"	"	"	"	-10 mA	2YP			"	2YP	"		"
	V_{OH2}	3006	6	-40 mA	1ZP			0.8 V	"	"	"	"	"	"					"	1ZP	2.0		"
			7	-40 mA	1ZP			2.0 V	0.8 V	"	"	"	"	"					"	1ZP	"		"
			8					"	2.0 V	"	"	"	"	0.8 V			2ZP	-40 mA	"	2ZP	"		"
			9			1YP	-40 mA	"	"	"	"	"	"	2.0 V			"	"	"	1YP	"		"
			10					"	"	"	"	"	"	"	-40 mA	2YP			"	2YP	"		"
	V_{OL}	3007	11			40 mA	1Y	0.8 V	"	"	"	"	"	"					"	1Y		0.4	"
			12			40 mA	1Y	2.0 V	0.8 V	"	"	"	"	"					"	1Y		"	"
			13					"	2.0 V	"	"	"	"	0.8 V	2Y	40 mA			"	2Y		"	"
			14	1Z	40 mA			"	"	"	"	"	"	2.0 V			40 mA	2Z	"	1Z		"	"
			15					"	"	"	"	"	"	2.0 V					"	2Z		"	"
	V_{IC}		16					-12 mA			"								"	1A		-1.5	"
			17						-12 mA		"								"	1B		"	"
			18							-12 mA	"								"	1C		"	"
			19								"	-12 mA							"	D		"	"
			20								"		-12 mA						"	2C		"	"
			21								"			-12 mA					"	2A		"	"
	V_{OC}		22		-40 mA						"	0.8 V							5.5 V	1Z		"	"
			23								"	"					-40 mA		"	2Z		"	"
			24			-40 mA					"	"							"	1Y		"	"
			25								"	"				-40 mA			"	2Y		"	"
	I_{IL1}	3009	26					0.4 V	5.5 V	5.5 V	"		5.5 V	5.5 V					"	1A	-0.7	-1.6	mA
			27					5.5 V	0.4 V	5.5 V	"		"	"					"	1B	"	"	"
			28					"	5.5 V	0.4 V	"		"	"					"	1C	"	"	"
			29					"	"	5.5 V	"		0.4 V	"					"	2C	"	"	"
			30					"	"	"	"		5.5 V	0.4 V					"	2A	"	"	"
	I_{IL2}	3009	31							"	"	0.4 V	5.5 V						"	D	-1.4	-3.2	"
	I_{IH1}	3010	32					2.4 V	GND	GND	"		GND	GND					"	1A		40	µA
			33					GND	2.4 V	GND	"		"	"					"	1B		"	"
			34					"	GND	2.4 V	"		"	"					"	1C		"	"
			35					"	"	GND	"		2.4 V	"					"	2C		"	"
			36					"	"	"	"		GND	2.4 V					"	2A		"	"
	I_{IH2}	3010	37					5.5 V	"	"	"		"	GND					"	1A		100	"
			38					GND	5.5 V	"	"		"	"					"	1B		"	"
			39					"	GND	5.5 V	"		"	"					"	1C		"	"
			40					"	"	GND	"		5.5 V	"					"	2C		"	"
			41					"	"	GND	"		GND	5.5 V					"	2A		"	"
	I_{IH3}	3010	42								"	2.4 V							"	D		80	"
	I_{IH4}	3010	43								"	5.5 V							"	D		200	"
	I_{O1}		44		12 V			0.8 V	2.0 V	2.0 V	"	2.0 V	2.0 V						"	1Z		100	"
			45		12 V			2.0 V	0.8 V	"	"	"	"						"	1Z		"	"
			46			12 V		"	2.0 V	"	"	"	"						"	1Y		"	"
			47					"	"	"	"	"	"	0.8 V			12 V		"	2Z		"	"
			48					"	"	"	"	"	"	2.0 V		12 V			"	2Y		"	"

'ee footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F Test no.	1 1ZP	2 1Z	3 1Y	4 1YP	5 1A	6 1B	7 1C	8 GND	9 D	10 2C	11 2A	12 2YP	13 2Y	14 2Z	15 2ZP	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_A = +25°C	I_{O2}		49	1Z	0.0 V					0.8 V	GND	2.0 V							5.5 V	1Z		-60	uA
			50			0.0 V	1Y			0.8 V	"	2.0 V							"	1Y		"	
			51	1Z	0.0 V					2.0 V	"	0.8 V							"	1Z		"	
			52			0.0 V	1Y			2.0 V	"	0.8 V							"	1Y		"	
			53								"	2.0 V	0.8 V				0.0 V	2Z	"	2Z		"	
			54								"	2.0 V	0.8 V		2Y	0.0 V			"	2Y		"	
			55								"	0.8 V	2.0 V				0.0 V	2Z	"	2Z		"	
			56								"	0.8 V	2.0 V		2Y	0.0 V			"	2Y		"	
	I_{O3}		57	1Z	0.4 V					0.8 V	"	2.0 V							"	1Z		440	"
			58			0.4 V	1Y			0.8 V	"	2.0 V							"	1Y		"	
			59	1Z	0.4 V					2.0 V	"	0.8 V							"	1Z		"	
			60			0.4 V	1Y			2.0 V	"	0.8 V							"	1Y		"	
			61								"	2.0 V	0.8 V				0.4 V	2Z	"	2Z		"	
			62								"	2.0 V	0.8 V		2Y	0.4 V			"	2Y		"	
			63								"	0.8 V	2.0 V				0.4 V	2Z	"	2Z		"	
			64								"	0.8 V	2.0 V		2Y	0.4 V			"	2Y		"	
	I_{O4}		65	1Z	2.4 V					0.8 V	"	2.0 V							"	1Z		"	
			66			2.4 V	1Y			0.8 V	"	2.0 V							"	1Y		"	
			67	1Z	2.4 V					2.0 V	"	0.8 V							"	1Z		"	
			68			2.4 V	1Y			2.0 V	"	0.8 V							"	1Y		"	
			69								"	2.0 V	0.8 V				2.4 V	2Z	"	2Z		"	
			70								"	2.0 V	0.8 V		2Y	2.4 V			"	2Y		"	
			71								"	0.8 V	2.0 V				2.4 V	2Z	"	2Z		"	
			72								"	0.8 V	2.0 V		2Y	2.4 V			"	2Y		"	
	I_{O5}		73	1Z	5.0 V					0.8 V	"	2.0 V							"	1Z		100	"
			74			5.0 V	1Y			0.8 V	"	2.0 V							"	1Y		"	
			75	1Z	5.0 V					2.0 V	"	0.8 V							"	1Z		"	
			76			5.0 V	1Y			2.0 V	"	0.8 V							"	1Y		"	
			77								"	2.0 V	0.8 V				5.0 V	2Z	"	2Z		"	
			78								"	2.0 V	0.8 V		2Y	5.0 V			"	2Y		"	
			79								"	0.8 V	2.0 V				5.0 V	2Z	"	2Z		"	
			80								"	0.8 V	2.0 V		2Y	5.0 V			"	2Y		"	
	I_{OS}	3011	81	0.0 V	1ZP			0.8 V	2.0 V	2.0 V	"	2.0 V							"	1ZP	-40	-120	mA
			82	0.0 V	1ZP			2.0 V	0.8 V	"	"	"							"	1ZP	"	"	
			83			1YP	0.0 V	2.0 V	2.0 V	"	"	"							"	1YP	"	"	
			84								"	"	2.0 V	0.8 V			2ZP	0.0 V	"	2ZP	"	"	
			85								"	"	2.0 V	2.0 V	0.0 V	2YP			"	2YP	"	"	
	I_{CC1}	3005	86					0.0 V	0.0 V	0.0 V	"	0.0 V	0.0 V	0.0 V					"	V_{CC}		65	"
	I_{CC2}	3005	87					0.0 V	0.0 V	0.0 V	"	0.0 V	0.0 V	0.0 V					7.0 V	V_{CC}		85	"
2 T_A = +125°C	V_{OH1}	3006	88	-10 mA	1ZP			0.8 V	2.0 V	2.0 V	"	2.0 V	2.0 V	2.0 V					4.5 V	1ZP	2.4		V
			89	-10 mA	1ZP			2.0 V	0.8 V	"	"	"	"	2.0 V					"	1ZP	"		
			90					"	2.0 V	"	"	"	"	0.8 V			2ZP	-10 mA	"	2ZP	"		
			91			1YP	-10 mA	"	"	"	"	"	"	2.0 V					"	1YP	"		
			92					"	"	"	"	"	"	"	-10 mA	2YP			"	2YP	"		
	V_{OL}	3007	93			40 mA	1Y	0.8 V	"	"	"	"	"	"					"	1Y		0.4	"
			94			40 mA	1Y	2.0 V	0.8 V	"	"	"	"	"					"	1Y		"	
			95					"	2.0 V	"	"	"	"	0.8 V	2Y	40 mA			"	2Y		"	
			96	1Z	40 mA			"	"	"	"	"	"	2.0 V					"	1Z		"	
			97					"	"	"	"	"	"	2.0 V			40 mA	2Z	"	2Z		"	

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F Test no.	1 1ZP	2 1Z	3 1Y	4 1YP	5 1A	6 1B	7 1C	8 GND	9 D	10 2C	11 2A	12 2YP	13 2Y	14 2Z	15 2ZP	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
2 T_A = +125°C	V_{OC}		98		-40 µA						GND	0.8 V					-40 mA		5.5 V	1Z		-1.5	V
			99								"	"							"	2Z		"	"
			100			-40 mA					"	"							"	1Y		"	"
			101								"	"				-40 mA			"	2Y		"	"
	I_{IL1}	3009	102					0.4 V			"								"	1A	-0.7	-1.6	mA
			103						0.4 V		"								"	1B	"	"	"
			104							0.4 V	"								"	1C	"	"	"
			105								"		0.4 V						"	2C	"	"	"
			106								"			0.4 V					"	2A	"	"	"
	I_{IL2}	3009	107								"	0.4 V							"	D	-1.4	-3.2	"
	I_{IH1}	3010	108					2.4 V			"								"	1A		40	µA
			109						2.4 V		"								"	1B		"	"
			110							2.4 V	"								"	1C		"	"
			111								"		2.4 V						"	2C		"	"
			112								"			2.4 V					"	2A		"	"
	I_{IH2}	3010	113					5.5 V			"								"	1A		100	"
			114						5.5 V		"								"	1B		"	"
			115							5.5 V	"								"	1C		"	"
			116								"		5.5 V						"	2C		"	"
			117								"			5.5 V					"	2A		"	"
	I_{IH3}	3010	118								"	2.4 V							"	D		80	"
	I_{IH4}	3010	119								"	5.5 V							"	D		200	"
	I_{O1}		120		12 V			0.8 V	2.0 V	2.0 V	"	2.0 V	2.0 V						"	1Z		"	"
			121		12 V			2.0 V	0.8 V	"	"	"	"						"	1Z		"	"
			122			12 V		"	2.0 V	"	"	"	"						"	1Y		"	"
			123					"	"	"	"	"	"	0.8 V			12 V		"	2Z		"	"
			124					"	"	"	"	"	"	2.0 V		12 V			"	2Y		"	"
	I_{O2}		125	1Z	0.0 V					0.8 V	"	"							"	1Z		-120	"
			126			0.0 V	1Y			0.8 V	"	"							"	1Y		"	"
			127	1Z	0.0 V					2.0 V	"	0.8 V							"	1Z		"	"
			128			0.0 V	1Y			2.0 V	"	0.8 V							"	1Y		"	"
			129								"	2.0 V	0.8 V				0.0 V	2Z	"	2Z		"	"
			130								"	2.0 V	0.8 V		2Y	0.0 V			"	2Y		"	"
			131								"	0.8 V	2.0 V				0.0 V	2Z	"	2Z		"	"
			132								"	0.8 V	2.0 V		2Y	0.0 V			"	2Y		"	"
	I_{O3}		133	1Z	0.4 V					0.8 V	"	2.0 V							"	1Z		+80	"
			134			0.4 V	1Y			0.8 V	"	2.0 V							"	1Y		"	"
			135	1Z	0.4 V					2.0 V	"	0.8 V							"	1Z		"	"
			136			0.4 V	1Y			2.0 V	"	0.8 V							"	1Y		"	"
			137								"	2.0 V	0.8 V				0.4 V	2Z	"	2Z		"	"
			138								"	2.0 V	0.8 V		2Y	0.4 V			"	2Y		"	"
			139								"	0.8 V	2.0 V				0.4 V	2Z	"	2Z		"	"
			140								"	0.8 V	2.0 V		2Y	0.4 V			"	2Y		"	"
	I_{O4}		141	1Z	2.4 V					0.8 V	"	2.0 V							"	1Z		"	"
			142			2.4 V	1Y			0.8 V	"	2.0 V							"	1Y		"	"
			143	1Z	2.4 V					2.0 V	"	0.8 V							"	1Z		"	"
			144			2.4 V	1Y			2.0 V	"	0.8 V							"	1Y		"	"
			145								"	2.0 V	0.8 V				2.4 V	2Z	"	2Z		"	"
			146								"	2.0 V	0.8 V		2Y	2.4 V			"	2Y		"	"
			147								"	0.8 V	2.0 V				2.4 V	2Z	"	2Z		"	"
			148								"	0.8 V	2.0 V		2Y	2.4 V			"	2Y		"	"

footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F Test no.	1 1ZP	2 1Z	3 1Y	4 1YP	5 1A	6 1B	7 1C	8 GND	9 D	10 2C	11 2A	12 2YP	13 2Y	14 2Z	15 2ZP	16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
2 T_A = +125°C	I_{OS}		149	1Z	5.0 V					0.8 V	GND	2.0 V							5.5 V	1Z		200	mA
			150			5.0 V	1Y			0.8 V	"	2.0 V							"	1Y		"	"
			151	1Z	5.0 V					2.0 V	"	0.8 V							"	1Z		"	"
			152			5.0 V	1Y			2.0 V	"	0.8 V							"	1Y		"	"
			153								"	2.0 V	0.8 V				5.0 V	2Z	"	2Z		"	"
			154								"	2.0 V	0.8 V		2Y	5.0 V			"	2Y		"	"
			155								"	0.8 V	2.0 V				5.0 V	2Z	"	2Z		"	"
			156								"	0.8 V	2.0 V		2Y	5.0 V			"	2Y		"	"
3	Same tests, terminal conditions, and limits as for subgroup 2, except T_A = -55°C.																						
9 T_A = +25°C 2/ 3/	t_{PLH1}	3003 (Fig. 4)	157	OUT	1ZP			IN	1/	1/	GND	1/							5.0 V	1ZP		25	ns
	t_{PHL1}		158	"	"			IN	1/	"	"	"							"	1ZP		"	"
	t_{PLH1}		159	"	"			1/	1W	"	"	"							"	1ZP		"	"
	t_{PHL1}		160					1/	IN		"	"					2ZP	OUT	"	2ZP		"	"
	t_{PLH1}		161								"	"	1/	IN			2ZP	OUT	"	2ZP		"	"
	t_{PHL1}		162								"	"	1/	IN					"	1YP		"	"
	t_{PLH2}		163			1YP	OUT	IN	1/	1/	"	"							"	1YP		"	"
	t_{PHL2}		164			"	"	IN	1/	"	"	"							"	1YP		"	"
	t_{PLH2}		165			"	"	1/	1W	"	"	"							"	2YP		"	"
	t_{PHL2}		166			"	"	1/	IN	"	"	"							"	2YP		"	"
	t_{PLH2}		167								"	"	1/	IN	OUT	2YP			"	2YP		"	"
	t_{PHL2}		168								"	"	1/	IN	OUT	2YP			"	2YP		"	"
	t_{PZH}		169	OUT	1ZP			GND	1/	IN	"	"							"	1ZP		20	"
	t_{PHZ}		170	"	"			"	"	IN	"	"							"	1ZP		30	"
	t_{PZH}		171	"	"			"	"	1/	"	"							"	1ZP		20	"
	t_{PHZ}		172	"	"			"	"	I/	"	"							"	1ZP		30	"
	t_{PZH}		173			1YP	OUT	1/	"	IW	"	1/							"	1YP		20	"
	t_{PHZ}		174			"	"	"	"	IN	"	I/							"	1YP		30	"
	t_{PZH}		175			"	"	"	"	1/	"	IW							"	1YP		20	"
	t_{PHZ}		176							I/	"	IN							"	1YP		30	"
	t_{PZH}		177								"	1/	IN	GND			2ZP	OUT	"	2ZP		20	"
	t_{PHZ}		178								"	I/	IN	"			"	"	"	2ZP		30	"
	t_{PZH}		179								"	IW	1/	"					"	2ZP		20	"
	t_{PHZ}		180								"	IN	I/	"					"	2YP		30	"
	t_{PZH}		181								"	1/	IN	1/	OUT	2YP			"	2YP		20	"
	t_{PHZ}		182								"	I/	IN	"	"	"			"	2YP		30	"
	t_{PZH}		183								"	IW	1/	"					"	2YP		30	"
	t_{PHZ}		184								"	IN	I/	"					"	2YP		30	"
	t_{PZL}		185	OUT	1ZP			1/	1/	IN	"	1/							"	1ZP		35	"
	t_{PLZ}		186	"	"			"	"	IN	"	I/							"	1ZP		30	"
	t_{PZL}		187	"	"			"	"	1/	"	IW							"	1ZP		35	"
	t_{PLZ}		188							I/	"	IN							"	1YP		30	"
	t_{PZL}		189			1YP	OUT	GND	"	IW	"	1/							"	1YP		35	"
	t_{PLZ}		190			"	"	"	"	IN	"	I/							"	1YP		30	"
	t_{PZL}		191			"	"	"	"	1/	"	IW							"	1YP		35	"
	t_{PLZ}		192							I/	"	IN							"	1YP		30	"
	t_{PZL}		193								"	1/	IN	1/			2ZP	OUT	"	2ZP		35	"
	t_{PLZ}		194								"	I/	IN	"			"	"	"	2ZP		30	"
	t_{PZL}		195								"	IW	1/	"					"	2ZP		35	"
	t_{PLZ}		196								"	IN	I/	"					"	2YP		30	"
	t_{PZL}		197								"	1/	IW	GND	OUT	2YP			"	2YP		35	"
	t_{PLZ}		198								"	I/	IN	"	"	"			"	2YP		30	"
	t_{PZL}		199								"	IW	1/	"					"	2YP		35	"
	t_{PLZ}		200								"	IN	I/	"					"				
10 2/ 3/	Same tests and terminal conditions as subgroup 9 with T_A = +125°C and the limits changed to: t_{PLH} and t_{PHL} now 30 ns maximum, t_{PZH} now 25 ns maximum, t_{PHZ} now 35 ns maximum, t_{PZL} now 35 ns maximum, t_{PLZ} now 45 ns maximum.																						
11 2/ 3/	Same tests, terminal conditions, and limits as subgroups 10, except T_A = -55°C.																						

1/ Indicated inputs are connected together to a 1 kΩ resistor to 5.0 V dc.
2/ For subgroups 9, 10, and 11, the unused output terminals may be open or loaded to facilitate equipment setup and minimize switching of loads as long as there is no effect on the measured parameter.
3/ For subgroups 9, 10, and 11, all the unused input terminals may be open or grounded at the supplier's option.

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TABLE III. Group A inspection for device types 06 and 07. 1/
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F Test no.	1 E_{B1}	2 E_{B2}	3 B2 OUT	4 B2 IN	5 B1 OUT	6 B1 IN	7 O/SE 1	8 GND	9 O/SE 2	10 A1 IN	11 A1 OUT	12 A2 IN	13 A2 OUT	14 EA_1	15 EA_2	16 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_A = +25°C	V_{OH}	3006	1							0.8 V	GND	0.8 V	2.0 V	-2 mA			0.8 V	0.8 V	4.5 V	11	2.4		V
			2							"	"	0.8 V			2.0 V	-2 mA	"	"	"	13	"		"
			3							"	"	2.0 V	0.8 V	-2 mA			"	"	"	11	"		"
			4							"	"	2.0 V			2.0 V	-2 mA	"	"	"	13	"		"
			5							2.0 V	"	0.8 V	0.8 V	-2 mA			"	"	"	11	"		"
			6							"	"	0.8 V			2.0 V	-2 mA	"	"	"	13	"		"
			7							"	"	2.0 V	0.8 V	-2 mA			"	"	"	11	"		"
			8							"	"	2.0 V			2.0 V	-2 mA	"	"	"	13	"		"
			9	0.8 V	0.8 V	-2 mA	2.0 V	-2 mA	2.0 V	0.8 V	"	0.8 V								5			
			10	"	"	-2 mA	2.0 V	-2 mA	2.0 V	"	"	0.8 V								3			
			11	"	"	-2 mA	2.0 V	-2 mA	2.0 V	"	"	2.0 V								5			
			12	"	"	-2 mA	2.0 V	-2 mA	2.0 V	2.0 V	"	2.0 V								3			
			13	"	"	-2 mA	2.0 V	-2 mA	2.0 V	"	"	2.0 V								5			
			14	"	"	-2 mA	2.0 V	-2 mA	2.0 V	"	"	0.8 V								3			
			15	"	"					"	"	0.8 V								5			
			16	"	"					"	"	2.0 V								3			
	V_{OL}	3007	17							0.8 V	GND	0.8 V	0.8 V	40 mA			0.8 V	0.8 V	4.5 V	11		0.5	
			18							"	"	0.8 V	2.0 V	40 mA	0.8 V	40 mA	"	"	"	13		"	
			19							"	"	2.0 V					"	"	"	11		"	
			20							"	"	2.0 V	2.0 V	40 mA	0.8 V	40 mA	"	"	"	13		"	
			21							2.0 V	"	0.8 V					"	"	"	11		"	
			22							"	"	0.8 V	2.0 V	40 mA	0.8 V	40 mA	"	"	"	13		"	
			23							"	"	2.0 V								5			
			24							"	"	2.0 V			0.8 V	40 mA				3			
			25	0.8 V	0.8 V			-2 mA	0.8 V	0.8 V	"	0.8 V								5			
			26	"	"	40 mA	0.8 V	-2 mA	2.0 V	"	"	0.8 V								3			
			27	"	"	40 mA	0.8 V	-2 mA	2.0 V	"	"	2.0 V								5			
			28	"	"	40 mA	0.8 V	-2 mA	2.0 V	2.0 V	"	2.0 V								3			
			29	"	"	40 mA	0.8 V	-2 mA	2.0 V	"	"	0.8 V								5			
			30	"	"	40 mA	0.8 V	-2 mA	2.0 V	"	"	0.8 V								3			
			31	"	"					"	"	2.0 V								5			
			32	"	"					"	"	2.0 V								3			
	V_{IC}		33	-12 mA																1			
			34		-12 mA															2			
			35				-12 mA													4			
			36						-12 mA											6			
			37							-12 mA										7			
			38									-12 mA								9			
			39										-12 mA							10			
			40												-12 mA					12			
			41														-12 mA			14			
			42															-12 mA		15			
	I_{IL}	3009	43	0.4 V	5.5 V		5.5 V		5.5 V	5.5 V		5.5 V	5.5 V		5.5 V		5.5 V	5.5 V	5.5 V	1	-0.7	-1.6	mA
			44	5.5 V	0.4 V		5.5 V		"	"		"	"		"		"	"	"	2	"	"	"
			45	"	5.5 V		0.4 V		0.4 V	"		"	"		"		"	"	"	4	"	"	"
			46	"	"		5.5 V		5.5 V	0.4 V		"	"		"		"	"	"	6	"	"	"
			47	"	"		"		"	5.5 V		"	"		"		"	"	"	7	"	"	"
			48	"	"		"		"	"		0.4 V	"		"		"	"	"	9	"	"	"
			49	"	"		"		"	"		5.5 V	0.4 V		"		"	"	"	10	"	"	"
			50	"	"		"		"	"		"	5.5 V		0.4 V		"	"	"	12	"	"	"
			51	"	"		"		"	"		"	"		5.5 V		0.4 V	"	"	14	"	"	"
			52	"	"		"		"	"		"	"		"		5.5 V	0.4 V	"	15	"	"	"

See footnote at end of device types 06 and 07.

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TABLE III. Group A inspection for device types 06 and 07 - Continued. 1/
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Test no.	E_{B1}	E_{B2}	B2 OUT	B2 IN	B1 OUT	B1 IN	D/SE 1	GND	D/SE 2	A1 IN	A1 OUT	A2 IN	A2 OUT	EA_1	EA_2	V_{CC}		Min	Max	
1 T_A = +25°C	I_{IH1}	3010	53	2.4 V	GND		GND		GND	GND	GND	GND	GND		GND		GND	GND	5.5 V	1		40	µA
			54	GND	2.4 V		GND		"	"	"	"	"		"		"	"	"	2		"	"
			55	"	GND		2.4 V		2.4 V	"	"	"	"		"		"	"	"	4		"	"
			56	"	"		GND		GND	2.4 V	"	"	"		"		"	"	"	6		"	"
			57	"	"		"		"	GND	"	"	"		"		"	"	"	7		"	"
			58	"	"		"		"	"	"	"	"		"		"	"	"	9		"	"
			59	"	"		"		"	"	"	2.4 V	"		"		"	"	"	10		"	"
			60	"	"		"		"	"	"	GND	2.4 V		"		"	"	"	12		"	"
			61	"	"		"		"	"	"	"	GND		2.4 V		"	"	"	14		"	"
			62	"	"		"		"	"	"	"	"		GND		2.4 V	2.4 V	"	15		"	"
	I_{IH2}	3010	63	5.5 V	GND		GND		GND	GND	"	GND	GND		GND		GND	GND	"	1		100	"
			64	GND	5.5 V		GND		"	"	"	"	"		"		"	"	"	2		"	"
			65	"	GND		5.5 V		5.5 V	"	"	"	"		"		"	"	"	4		"	"
			66	"	"		GND		GND	5.5 V	"	"	"		"		"	"	"	6		"	"
			67	"	"		"		"	GND	"	"	"		"		"	"	"	7		"	"
			68	"	"		"		"	"	"	"	"		"		"	"	"	9		"	"
			69	"	"		"		"	"	"	5.5 V	"		"		"	"	"	10		"	"
			70	"	"		"		"	"	"	GND	5.5 V		5.5 V		"	"	"	12		"	"
			71	"	"		"		"	"	"	"	GND		GND		"	"	"	14		"	"
			72	"	"		"		"	"	"	"	"		GND		5.5 V	5.5 V	"	15		"	"
	I_{OS}	3011	73	0.8 V	0.8 V	GND	4.5 V	GND	4.5 V	0.8 V	"	0.8 V	4.5 V	GND	4.5 V	GND	0.8 V	0.8 V	"	3	-40	-120	mA
			74	"	"					"	"	"					"	"	"	5	"	"	"
			75	"	"					"	"	"					"	"	"	11	"	"	"
			76	"	"					"	"	"					"	"	"	13	"	"	"
	I_{CC}	3005	77	GND	GND		GND		GND	GND	"	GND	GND		GND		GND	GND	"	16		90	"
	V_{OC}		78			-12 mA		-12 mA			"			-12 mA	-12 mA				"	3		-1.5	V
			79								"								"	5		"	"
			80								"								"	11		"	"
			81								"								"	13		"	"
	I_{OI}		82							0.8 V	"	0.8 V	2.0 V	2.4 V	2.0 V	2.4 V	2.0 V	0.8 V	"	11	-120	40	µA
			83							"	"	"	2.0 V	2.4 V	"	"	2.0 V	0.8 V	"	13	"	"	"
			84							2.0 V	"	2.0 V	0.8 V	2.4 V	2.0 V	2.4 V	0.8 V	2.0 V	"	11	"	"	"
			85							"	"	"	0.8 V	2.4 V	"	"	0.8 V	0.8 V	"	13	"	"	"
			86							"	"	"	"	"	2.0 V	2.4 V	0.8 V	2.0 V	"	11	"	"	"
			87							"	"	"	"	"	"	"	0.8 V	0.8 V	"	13	"	"	"
			88							0.8 V	"	0.8 V	"	"	2.0 V	2.4 V	0.8 V	2.0 V	"	11	"	"	"
			89							"	"	"	"	"	"	"	"	"	"	13	"	"	"
			90	2.0 V	0.8 V	2.4 V	2.0 V	2.4 V	2.0 V	"	"	0.8 V	"	"	"	"	"	"	"	5	"	"	"
			91	2.0 V	0.8 V	"	"	2.4 V	2.0 V	"	"	"	"	"	"	"	"	"	"	3	"	"	"
			92	0.8 V	2.0 V	2.4 V	2.0 V	"	"	"	"	"	"	"	"	"	"	"	"	5	"	"	"
			93	0.8 V	2.0 V	"	"	2.4 V	0.8 V	2.0 V	"	2.0 V	"	"	"	"	"	"	"	3	"	"	"
			94	2.0 V	0.8 V	2.4 V	2.0 V	"	"	"	"	"	"	"	"	"	"	"	"	5	"	"	"
			95	2.0 V	0.8 V	"	"	2.4 V	"	"	"	"	"	"	"	"	"	"	"	3	"	"	"
			96	0.8 V	2.0 V	2.4 V	2.0 V	"	"	"	"	"	"	"	"	"	"	"	"	5	"	"	"
			97	0.8 V	2.0 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	3	"	"	"

See footnote at end of device types 06 and 07.

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TABLE III. Group A inspection for device types 06 and 07 - Continued. 1/
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Test no.	E_{B1}	E_{B2}	B2 OUT	B2 IN	B1 OUT	B1 IN	D/SE 1	GND	D/SE 2	A1 IN	A1 OUT	A2 IN	A2 OUT	EA_1	EA_2	V_{CC}		Min	Max	
1 T_A = +25°C	I_{O2}		98							0.8 V	GND	0.8 V	0.8 V	0.4 V			2.0 V	0.8 V	5.5 V	11	-160	40	µA
			99							"	"	"			0.8 V	0.4 V	2.0 V	0.8 V	"	13	"	"	"
			100							"	"	"	0.8 V	0.4 V			0.8 V	2.0 V	"	11	"	"	"
			101							"	"	"			0.8 V	0.4 V	0.8 V	2.0 V	"	13	"	"	"
			102							2.0 V	"	2.0 V	2.0 V	0.4 V			2.0 V	0.8 V	"	11	"	"	"
			103							"	"	"			0.8 V	0.4 V	2.0 V	0.8 V	"	13	"	"	"
			104							"	"	"	2.0 V	0.4 V			0.8 V	2.0 V	"	11	"	"	"
			105							"	"	"			0.8 V	0.4 V	0.8 V	2.0 V	"	13	"	"	"
			106	2.0 V	0.8 V			0.4 V	0.8 V	0.8 V	"	0.8 V							"	5	"	"	"
			107	2.0 V	0.8 V	0.4 V	0.8 V			"	"	"							"	3	"	"	"
			108	0.8 V	2.0 V			0.4 V	0.8 V	"	"	"							"	5	"	"	"
			109	0.8 V	2.0 V	0.4 V	0.8 V			"	"	"							"	3	"	"	"
			110	2.0 V	2.0 V			0.4 V	2.0 V	2.0 V	"	2.0 V							"	5	"	"	"
			111	2.0 V	0.8 V	0.4 V	0.8 V			"	"	"							"	3	"	"	"
			112	0.8 V	2.0 V			0.4 V	2.0 V	"	"	"							"	5	"	"	"
			113	0.8 V	2.0 V	0.4 V	0.8 V			"	"	"							"	3	"	"	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_A = +125°C, V_{IC} and V_{OC} are omitted. V_{IL} = 0.7 V .																						
3	Same tests, terminal conditions, and limits as subgroup 7, except T_A = -55°C, V_{IC} and V_{OC} are omitted.																						
9 T_A = +25°C	t_{PLH}	3003 (Fig. 4)	114							5.0 V	GND	GND	IN	OUT			GND	GND	5.0 V	11	2	25	ns
			115							"	"	"			IN	OUT	GND	GND	"	13	"	"	"
			116	GND	GND	OUT	IN	OUT	IN	"	"	"					GND	GND	"	5	"	"	"
			117	GND	GND					"	"	"					GND	GND	"	3	"	"	"
	t_{PLH}		118							"	"	"	IN	OUT			GND	GND	"	11	"	"	"
			119							"	"	"			IN	OUT	GND	GND	"	13	"	"	"
			120	GND	GND	OUT	IN	OUT	IN	"	"	"					GND	GND	"	5	"	"	"
			121	GND	GND					"	"	"					GND	GND	"	3	"	"	"
	t_{PLH} t_{PLH}		122							GND	"	"	IN	OUT			GND	GND	"	11	"	"	"
			123	GND	GND			OUT	IN	"	"	"					GND	GND	"	5	"	"	"
	t_{PHL} t_{PHL}		124							"	"	"	IN	OUT			GND	GND	"	11	"	"	"
			125	GND	GND			OUT	IN	"	"	"					GND	GND	"	5	"	"	"
	t_{PZH}		126							5.0 V	"	"	GND	OUT			IN	GND	"	11	"	22	"
			127							"	"	"			5.0 V	OUT	IN	GND	"	13	"	"	"
			128	IN	GND	OUT	5.0 V	OUT	GND	"	"	"					GND	GND	"	5	"	"	"
			129	IN	GND					"	"	"					GND	GND	"	3	"	"	"
	t_{PZL}		130							"	"	"	5.0 V	OUT			IN	GND	"	11	"	25	"
			131							"	"	"			GND	OUT	IN	GND	"	13	"	"	"
			132	IN	GND	OUT	GND	OUT	5.0 V	"	"	"					GND	GND	"	5	"	"	"
			133	IN	GND					"	"	"					GND	GND	"	3	"	"	"
	t_{PHZ}		134							"	"	"	GND	OUT			IN	GND	"	11	"	32	"
			135							"	"	"			5.0 V	OUT	IN	GND	"	13	"	"	"
			136	IN	GND	OUT	5.0 V	OUT	GND	"	"	"					GND	GND	"	5	"	"	"
			137	IN	GND					"	"	"					GND	GND	"	3	"	"	"
	t_{PLZ}		138							"	"	"	5.0 V	OUT			IN	GND	"	11	"	27	"
			139							"	"	"			GND	OUT	IN	GND	"	13	"	"	"
			140	IN	GND	OUT	GND	OUT	5.0 V	"	"	"					GND	GND	"	5	"	"	"
			141	IN	GND					"	"	"					GND	GND	"	3	"	"	"

See footnote at end of device types 06 and 07.

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TABLE III. Group A inspection for device types 06 and 07 - Continued. 1/
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Cases E and F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Test no.	E_{B1}	E_{B2}	B2 OUT	B2 IN	B1 OUT	B1 IN	D/SE 1	GND	D/SE 2	A1 IN	A1 OUT	A2 IN	A2 OUT	EA_1	EA_2	V_{CC}		Min	Max	
10 T_A = +125°C	t_{PLH}	3003 (Fig. 4)	142 143 144 145	GND GND	GND GND	OUT	IN	OUT	IN	5.0 V " "	GND " "	GND " "	IN	OUT	IN	OUT	GND GND	GND GND	5.0 V " "	11 13 5 3	2 " "	28 " "	ns " "
	t_{PHL}		146 147 148 149	GND GND	GND GND	OUT	IN	OUT	IN	" " "	" " "	" " "	IN	OUT	IN	OUT	GND GND	GND GND	" " "	11 13 5 3	" " "	" " "	" " "
	t_{PLH} t_{PLH}		150 151	GND	GND			OUT	IN	GND "	" "	" "	IN	OUT			GND	GND	" "	11 5	" "	" "	" "
	t_{PHL} t_{PHL}		152 153	GND	GND			OUT	IN	" "	" "	" "	IN	OUT			GND	GND	" "	11 5	" "	" "	" "
	t_{PZH}		154 155 156 157	IN IN	GND GND	OUT	5.0 V	OUT	GND	5.0 V " "	" " "	" " "	GND	OUT	5.0 V	OUT	IN IN	GND GND	" " "	11 13 5 3	" " "	32 " "	" " "
	t_{PZL}		158 159 160 161	IN IN	GND GND	OUT	GND	OUT	5.0 V	" " "	" " "	" " "	5.0 V	OUT	GND	OUT	IN IN	GND GND	" " "	11 13 5 3	" " "	" " "	" " "
	t_{PHZ}		162 163 164 165	IN IN	GND GND	OUT	5.0 V	OUT	GND	" " "	" " "	" " "	GND	OUT	5.0 V	OUT	IN IN	GND GND	" " "	11 13 5 3	" " "	34 " "	" " "
	t_{PLZ}		166 167 168 169	IN IN	GND GND	OUT	GND	OUT	5.0 V	" " "	" " "	" " "	5.0 V	OUT	GND	OUT	IN IN	GND GND	" " "	11 13 5 3	" " "	36 " "	" " "
11	Same tests, terminal conditions as subgroup 10, except T_A = -55°C.																						

1/ IN = input impulse, OUT = pin under test.

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4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Life test for class B (method 1005 of MIL-STD-883). Test condition D for types 01 and 02, and test condition B for types 03, 04, 05, 06, and 07 using the circuit shown on figure 3.

c. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of examination and test. Methods of examination and test shall be as specified in the appropriate tables. Electrical test circuits as prescribed herein or in the referenced test methods of MIL-STD-883 shall be acceptable. Other test circuits shall require approval of the qualifying activity.

4.5.1 Voltage and current. All voltage values given are referenced to the ground terminal of the device under test (DUT). Current values given are conventional and are positive when flowing into the referenced terminal.

4.5.2 Life test cooldown procedures. When devices are measured at +25°C following application of the steady-state life or burn-in test condition, they shall be cooled to within 10°C of their power stable condition (at room temperature) prior to removal of the bias.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special lead lengths, or lead forming, if applicable. These requirements shall not affect the part number.

h. Requirements for "JAN" marking.

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6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Electrical ground (common terminal)
V_{IN}	Voltage level at an input terminal
I_{IN}	Current flowing into an input terminal
BV_{IN}	High level input breakdown voltage
I_{CEX}	Output leakage current
V_{IC}	Input clamp voltage
V_{OC}	Output clamp voltage
V_{TH}	Differential input threshold voltage
V_{CM}	Common-mode input voltage
t_{PHZ}	Output-disable time (of a three-state output) from high level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined high level to a high impedance (off) state.
t_{PLZ}	Output disable time (of a three-state output) from low level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from the defined low level to a high-impedance (off) state.
t_{PZH}	Output enable time (of a three-state output) to high level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high-impedance (off) state to the defined high level.
t_{PZL}	Output enable time (of a three-state output) to low level. The time between the specified reference points on the input and output voltage waveforms with the three-state output changing from a high-impedance (off) state to the defined low level.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	55107
02	55108
03	9614, 55114
04	9615, 55115
05	55113
06	7831
07	7832

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6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/104B	MICROCIRCUITS, LINEAR LINE DRIVERS AND RECEIVERS MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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